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EVALUATING SU(3) VERLINDE SUMS USING SPECTRAL GRAPH THEORY

JAY JORGENSON, ANDERS KARLSSON, AND LEJLA SMAJLOVIĆ

Abstract. We realize the SU(3) Verlinde sums $V_n(m)$, up to an explicit factor, as the values at n of the spectral zeta function of a higher-order Laplace operator on the triangular discrete torus on m^2 vertices. For fixed m, we express their generating function in terms of the logarithmic derivative of an associated even spectral polynomial and express this polynomial as an explicit iterated resultant. Exploiting permutation symmetry, we prove that this polynomial is a cube over $\mathbb{Q}$, apart from an explicit quadratic factor when $3 \mid m$. This factorization yields shorter linear recurrences satisfied by $V_n(m)$ with constant coefficients. We also derive Binet-type formulas expressing $V_n(m)$ as finite linear combinations of powers of rescaled inverse squares of the roots of the spectral polynomial. These results provide an efficient algorithm for computing these Verlinde sums. Several fully developed examples demonstrating computational efficiency of the method are given, including expressions in terms of Fibonacci and Lucas numbers.

1. Introduction

Verlinde's formula, originally proposed from conformal field theory, gives the dimension of a certain finite-dimensional vector space associated with a compact Lie group, the genus g of a Riemann surface, and a positive integral level k [Ve88]; see also the mathematical introduction to conformal field theory in [Sc08]. Its proofs and interpretation in terms of conformal blocks and generalized theta functions was developed in [BL94, TUY89]. Different proofs and algebro-geometric formulations were obtained in [Fa94, KNR94, Be96, Ku22]. Related approaches to Verlinde's formula include stable bundles and stable pairs [Bo91, Th94], the derivation of the G/G model from Chern–Simons theory [BT93], and a loop-group fixed point formula on the moduli space of flat bundles [AMW01].

In this paper, as in [Za96], we suppress the geometric, algebraic, topological and physical background and study directly the finite trigonometric expressions, which we call *Verlinde sums*.

Early combinatorial and residue theoretic work on the Verlinde formula includes Szenes's verification of the SU(2) case and his subsequent combinatorial treatment [Sz91, Sz95]. Residue and rational trigonometric sum methods were developed further in [Sz98, Sz03]; related residue formulas for vector partitions and Euler–Maclaurin sums appear in [SV03, BrVe97], while wall-crossing and partition-function techniques relevant to this framework appear in [BoVe09]. An interested reader may also consult [Ve03] for an exposition of residue formulas for Verlinde sums. More recently, Verlinde sums were treated as rational trigonometric sums in [LM22], with their relation to quantization considered in [LM22b]. Spectral and discrete-analytic methods for related trigonometric sums appear in [Do92, CHJSV23a, JKS24, KM26].

Dowker remarked in [Do92, p. 2642] that the SU(2) Verlinde sums are special values of the spectral zeta function of discrete circles and asked whether related sums admit a similar spectral interpretation. We answer this question for SU(3) by realizing the corresponding Verlinde sums as special values of the spectral zeta function of a higher-order Laplace operator on a triangular

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discrete torus. Whereas earlier residue and generating-function approaches [Sz95, Za96] study these sums principally as functions of the level for fixed genus, our spectral-polynomial approach addresses their genus dependence for fixed level.

This spectral realization leads to an explicit and effective algorithm for computing the SU(3) Verlinde sums for arbitrary fixed level and genus parameter; see Section 1.5. More precisely, we construct the associated spectral polynomial by means of an explicit iterated resultant and use its additional symmetries to reduce the computational complexity. The logarithmic derivative of the reduced polynomial yields a rational generating function, substantially shorter constant coefficient recurrences, and Binet-type formulas expressing the Verlinde sums as finite sums of powers of algebraic numbers. Consequently, the theoretical framework gives a symbolic computational procedure based on polynomial arithmetic, resultants, factorization, and recurrence evaluation. The fully worked examples demonstrate the effectiveness of the method even when the unreduced spectral polynomial has high degree and the resulting Verlinde sums are extremely large.

For example, we prove that the Verlinde sum $V_{g-1}(k+3)$ on SU(3) associated to the genus $g>1$ and level $k=2$ can be explicitly expressed in terms of Fibonacci numbers as

$$V_{g-1}(5)=\begin{cases} 2\cdot 3^g 5^{\frac{g}{2}} F_{g-1}, & \text{if } g \text{ is even} \\ 2\cdot 3^g 5^{\frac{g-1}{2}}(F_{g-2}+F_g), & \text{if } g \text{ is odd} \end{cases},$$

see equation (28) in the Appendix. Another example, from the expression (34) for the Verlinde sum $V_{g-1}(10)$, $n=g-1\geq 1$:

$$V_n(10)=\begin{cases} 6\cdot 5^{n/2}\left[(2\cdot 15^n+12^n+2\cdot 60^n)L_n+60^n L_{5n}\right], & n \text{ even}, \\ 6\cdot 5^{(n+1)/2}\left[(2\cdot 15^n+12^n+2\cdot 60^n)F_n+60^n F_{5n}\right], & n \text{ odd}, \end{cases} \tag{1}$$

where L_n denotes the nth Lucas number.

1.1. **Verlinde sums for** SU(2)**.** The study of Verlinde sums for SU(2) amounts to understanding the quantities

$$C_m(n):=\sum_{j=1}^{m-1}\frac{1}{(\sin(\pi j/m))^{2n}} \quad \text{for } n,m\in\mathbb{Z}_{>1}. \tag{2}$$

In [Za96] there are nine equivalent characterizations of these numbers. The quantities (2) were computed explicitly, in both untwisted and twisted forms, by Dowker [Do92], who evaluated the corresponding sums of powers of cosecants in closed form in connection with Verlinde's formula for the SU(2) Wess–Zumino–Witten model; these cosecant sums are precisely the object recast spectrally in [JKS24]. The point of view of [JKS24] adds to the content of [Do92, Za96] by realizing (2) using the spectrum of the discrete Laplacian acting on a certain function space on a finite graph. The graph in question is the discrete circle $\mathbb{Z}/m\mathbb{Z}$, and the function space is the set of functions on $\mathbb{Z}/m\mathbb{Z}$ (an additional twist is admissible). The setting is as follows.

For $r\in\{-(m-1),\dots,m-1\}$, let

$$C_{m,r}(n):=\frac{1}{m}\sum_{j=1}^{m-1}\csc^{2n}\Big(\frac{j}{m}\pi\Big)e^{2\pi i r j/m} \tag{3}$$

As proved in [JKS24], the corresponding generating functions

$$f_{m,r}(s)=\sum_{n\geq 0}C_{m,r}(n+1)\,s^n$$

are for sufficiently small s convergent series and are equal to rational functions which can be expressed explicitly in terms of Chebyshev polynomials T_m, U_m of the first and second kind. Specifically, it is shown in [JKS24] that

$$f_{m,r}(s) = 2\,\frac{U_{m-\ell-1}(1-2s) + U_{\ell-1}(1-2s)}{T_m(1-2s)-1} + \frac{1}{ms}, \tag{4}$$

where $\ell \in \{0,\dots,m-1\}$ satisfies $\ell \equiv r \pmod m$ and $U_{-1} \equiv 0$. From (4) one obtains both closed forms and a one-step recursion in m for (2).

1.2. **Verlinde sums for** SU(3) **and spectral graph theory.** Following [Za96], we take as the definition of the SU(3) Verlinde sum $V_n(m)$, for positive integers n and m, to be the series

$$V_n(m) = 3^n m^{2n} \sum_{\substack{(j,\ell,k)\in(\mathbb{Z}/m\mathbb{Z}\setminus\{0\})^3 \\ j+\ell+k\equiv 0 \bmod m}} \Big(8\sin\frac{\pi j}{m}\sin\frac{\pi \ell}{m}\sin\frac{\pi k}{m}\Big)^{-2n}. \tag{5}$$

When comparing our notation to [Za96], our n (resp. m) corresponds to $g-1$ (resp. $k+3$). In a slight abuse of terminology, one can refer to n as a genus and m as a level, noting here that, in actuality, $n = g-1$ and $m = k+3$. Therefore, unless otherwise stated, throughout this paper we assume $m \geq 3$.

Consider the standard lattice graph $\mathbb{Z}^3$ and its quotient torus $\mathrm{DT}(m) = (\mathbb{Z}/m\mathbb{Z})^3$, which is a three-dimensional discrete torus with edges in the coordinate directions. Take a further quotient by the free diagonal action

$$(x_1,x_2,x_3) \longmapsto (x_1+1, x_2+1, x_3+1).$$

The map

$$[x_1,x_2,x_3] \longmapsto (x_1-x_3, x_2-x_3)$$

identifies the quotient with

$$G_m = (\mathbb{Z}/m\mathbb{Z})^2.$$

Writing $x = (x_1,x_2)$, its six neighbors are

$$x \pm e_1, \qquad x \pm e_2, \qquad x \pm (e_1+e_2).$$

Thus G_m is what could be called an example of a *circulant torus*, or more specifically a six-regular triangular discrete torus on m^2 vertices. The quotient map $\mathrm{DT}(m) \to G_m$ is an m-to-one regular covering. See Figure 1.

For $d \in G_m$, let T_d denote the operator

$$(T_d f)(x) := f(x+d)$$

and define a directional Laplacian as follows:

$$\Delta_d = (T_d - I)^*(T_d - I) = 2I - T_d - T_{-d}.$$

The corresponding higher-order Laplace operator is

$$\Delta_m = \frac{1}{4}\Delta_{e_1}\Delta_{e_2}\Delta_{e_1+e_2}.$$

As will be explained below, for $j = (a,b)$, the eigenvalues of Δ_m are

$$\lambda_m(j) = 16\sin^2(\pi a/m)\sin^2(\pi b/m)\sin^2(\pi(a+b)/m).$$

This gives a spectral zeta interpretation of the Verlinde sums

$$V_n(m) = \big(3m^2/4\big)^n \zeta_{\Delta_m}(n).$$

Here the spectral zeta function is a sum over the non-zero eigenvalues, more precisely

$$\zeta_{\Delta_m}(s) = \sum_{j\in G_m\setminus Z} \lambda_m(j)^{-s}$$

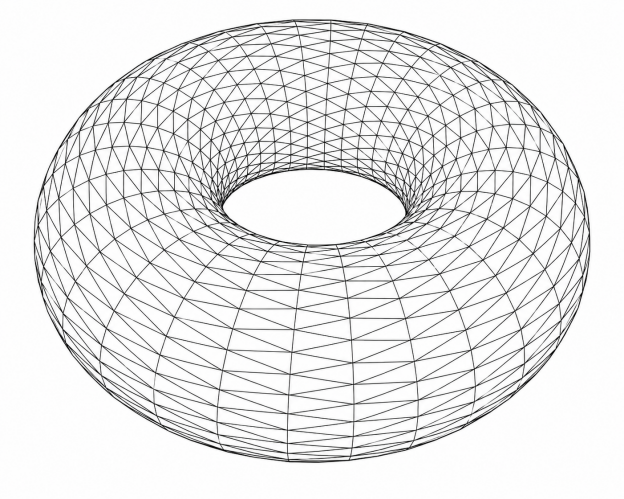


FIGURE 1. The two dimensional triangular torus G_m

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where $Z = \{j = (a, b) : a = 0, b = 0, \text{or } a + b \equiv 0\}$. [1]

From this construction, we state our first theorem which is the following.

Theorem 1.1. *Let G_m be the 2-dimensional discrete triangular torus on m^2 vertices and Δ_m defined above. Set $c_m := 3m^2/4$. Then*

$$V_n(m) = (3m^2/4)^n \zeta_{\Delta_m}(n).$$

Moreover, let $H(w) = \prod_{j \in G_m \setminus Z}(\lambda_m(j) - w)$ be the reduced spectral polynomial of Δ_m, and the rational function

$$F(w) = -\frac{1}{m^2}\frac{H'(w)}{H(w)}$$

Then

$$V_n(m) = 3^n \Big(\frac{m}{2}\Big)^{2n} \cdot \frac{m^2}{(n-1)!}\, \partial_w^{n-1} F(w)\Big|_{w=0}$$

and

$$\sum_{n \geq 1} V_n(m)\, \tau^n = c_m \tau \cdot m^2 F(c_m \tau). \tag{6}$$

as an identity of formal power series in τ near $\tau = 0$. Indeed, the series in (6) converges absolutely and uniformly on every closed disc $|\tau| \leq R < \frac{1}{c_m} \min_{j \notin Z} \lambda_m(j)$.

One can view the last part of Theorem 1.1 as the generalization to SU(3) of (4) from [JKS24].

1.3. **Verlinde sums for** SU(3) **and resultants.** In [JKS24] it was shown that the resolvent kernel $G(x, y; s)$ for $\mathbb{Z}/m\mathbb{Z}$ when $x = y$ can be given in terms of Chebyshev polynomials; see As a consequence, one was able to obtain linear recursion formulas for the series (2) and (3). The following theorem gives the analogous statement for SU(3). In this case, we replace the Chebyshev polynomials with the resultant $\mathrm{Res}_x(f, g)$ which is constructed from elementary and explicit polynomials which depend solely on m.

For the generating function statement below, set $V_0(m) := (m-1)(m-2)$.

[1] The complex function $z \mapsto z^{-s}$ for $z \in \mathbb{C} \setminus \{0\}$ is defined using the principal branch of the logarithm.

Theorem 1.2. *For any integer $m \geq 3$ and variables u, v and w, let*

$$Q(u) := u^m - i^m \quad \text{and} \quad W(u,v,w) := iu^2v^2 + u^2v + uv^2 - 2wuv + u + v - i, \tag{7}$$

where $i = \sqrt{-1}$. Let

$$D(w) = \frac{1}{2^{m^2} w^{3m-2}} \operatorname{Res}_u \Big(Q(u),\ \operatorname{Res}_v \big(Q(v),\ W(u,v,w) \big) \Big). \tag{8}$$

Then $D(w)$ is an even, monic polynomial in w of degree $(m-1)(m-2)$. Furthermore, for sufficiently small τ and fixed m, the generating series (6) for Verlinde sums satisfies the relation that

$$\sum_{n \geq 0} V_n(m)\, \tau^n = (m-1)(m-2) - \sqrt{c_m \tau}\, \frac{D'(\sqrt{c_m \tau})}{D(\sqrt{c_m \tau})} \quad \text{where} \quad c_m := \frac{3m^2}{4}. \tag{9}$$

The resultant in (8) is sufficiently constructive that it is amenable for explicit computations. Throughout this article we will give numerous examples for various m.

1.4. **Verlinde sums for** SU(3) **and sums of powers.** If one were to multiply both sides of (9) by D, one gets a recursive formula in n of length $\deg(D)/2 = (m-1)(m-2)/2$ for the Verlinde sums $V_n(m)$ for fixed m. (The degree is divided by 2, because the generating variable τ is such that $w^2 = c_m \tau$, while $\deg(D)$ signifies the degree of D in variable w.) The recursive formulas are obtained by equating coefficients in τ in the resulting product. However, because the right-hand-side of (9) is the logarithmic derivative of D, the length can be reduced whenever D has multiple roots, meaning D and D' have common factors. The following theorem shows that, in general, D is highly divisible, so then one can expect recursive formulas for $V_n(m)$ which are much shorter.

Theorem 1.3. *For every $m \geq 3$, there is a polynomial $E(w) \in \mathbb{Q}[w]$ such that*

$$D(w) = \begin{cases} E(w)^3, & 3 \nmid m, \\ \big(w^2 - \frac{27}{4}\big) E(w)^3, & 3 \mid m. \end{cases}$$

As such, $\deg E = (m-1)(m-2)/3$ *if* $3 \nmid m$ *and* $\deg E = ((m-1)(m-2)-2)/3$ *if* $3 \mid m$.

From Theorem 1.3, we have that

$$\frac{D'}{D} = \frac{3E'}{E} + \frac{2w}{w^2 - 27/4} \mathbf{1}_{3|m}$$

where $\mathbf{1}_{3|m} = 1$ if 3 divides m and zero otherwise, and the prime denotes the derivative in w. Therefore, Theorem 1.3 implies that the length of the recursive sequence is at most $\frac{\deg E}{2} + \mathbf{1}_{3|m}$. In most examples, the values attained by $V_n(m)$ are very large, so the reduction in the length of the recursive formula is significant when obtaining numerical evaluations.

There is one additional noteworthy aspect of Theorem 1.3, namely that in the generating series (9), the right-hand-side has simple poles. Therefore, from the elementary expansion

$$\frac{bx}{x-a} = -b \sum_{n=1}^{\infty} (x/a)^n \quad \text{for constants } a,\ b \text{ and } a \neq 0 \text{ and } |x| < |a|,$$

we get the following expression for Verlinde sums in terms of the roots of E.

Theorem 1.4. *With notation as above, let $\mathcal{E}_m = \{a\}$ be the set of all distinct roots of E, each with multiplicity $\mu(a)$. Then, for any $n \geq 1$, we have that*

$$V_n(m) \ = \ \mathbf{1}_{3|m} \cdot 2\Big(\frac{m}{3}\Big)^{2n} \ + \ 3 \sum_{a \in \mathcal{E}_m} \mu(a) \left(\frac{3m^2}{4a^2}\right)^n. \tag{10}$$

Theorem 1.4 is the analogue of the classical Binet formula for linear recurrence sequence. In effect, the factor of 2 in (10) is because of the change of variables from w to τ, and the additional factor of 3 in the sum over roots of E is from Theorem 1.3. Any additional appearance of repeated roots of E manifests itself through the factor $\mu(a)$. In the case $m = 10$ which we give in the Appendix below, the resultant D has degree 72, yet we obtain a Binet formula involving 4 pairs of algebraic integers in $\mathbb{Q}(\sqrt{5})$.

Alternatively, let $\widehat{E}(x) = E(w)|_{w^2=x}$ and define its reciprocal polynomial by

$$\widehat{E}^{\vee}(y) = y^{\deg \widehat{E}}\widehat{E}(1/y).$$

Let $\mathcal{E}^{\vee}$ be the set of distinct roots of $\widehat{E}^{\vee}$, with multiplicities $\mu(\alpha)$. Then

$$V_n(m) = \mathbf{1}_{3|m}\, 2\left(\frac{m}{3}\right)^{2n} + 6\cdot 3^n\left(\frac{m}{2}\right)^{2n}\sum_{\alpha\in\mathcal{E}^{\vee}}\mu(\alpha)\alpha^n. \tag{11}$$

1.5. **An algorithm for Verlinde sums for** SU(3)**.** When combining the results above, we obtain the following effective algorithm to explicitly compute $V_n(m)$.

Algorithm (Computation of $V_n(m)$).

INPUT: Integers $m \geq 3$ and $n \geq 1$.

1. Let $Q(u) := u^m - i^m$ and $W(u,v,w) := iu^2v^2 + u^2v + uv^2 - 2wuv + u + v - i$, and compute the iterated resultant
$$D(w) \;=\; \frac{1}{2^{m^2}w^{3m-2}}\,\mathrm{Res}_u\Big(Q(u),\ \mathrm{Res}_v\big(Q(v),\,W(u,v,w)\big)\Big).$$
2. Extract the reduced polynomial $E(w)$ and let $\widehat{E}(x) := E(w)|_{w^2=x}$.
3. Let $E^{\vee}(y) := y^{\deg\widehat{E}}\,\widehat{E}(1/y)$. Let $\mathcal{E}^{\vee}$ be the set of distinct roots of $E^{\vee}$ where each root α has multiplicity $\mu(\alpha)$. Set
$$p_n \;:=\; \sum_{\alpha\in\mathcal{E}^{\vee}}\mu(\alpha)\alpha^n.$$

OUTPUT: The Verlinde sum $V_n(m)$ is given by

$$V_n(m) \;=\; \mathbf{1}_{3|m}\cdot 2\Big(\frac{m}{3}\Big)^{2n} \;+\; 6\cdot 3^n\Big(\frac{m}{2}\Big)^{2n} p_n. \tag{12}$$

1.6. **Organization of this paper.** In section 2 we establish notation and recall background information. In particular, we discuss the spectral theory on the circulant torus, which is needed for Theorem 1.2, and basic facts for the resultant and linear recurrences, which is needed for the other results stated above. In section 3 we prove Theorem 1.1. In Section 4 we prove the first resultant construction on G_m, present a few examples, and discuss its inefficiency. Our second resultant construction is given in Section 5, namely formula (8). Further examples are given which demonstrate the improvement over the resultant construction from Section 4. In Section 6 we complete the proof of Theorem 1.2 and derive, from the rational generating function, the linear recurrences for $V_n(m)$; these use the function D from (8). In Section 7 we prove Theorem 1.3 which allows one to repeat the analysis from 4 and prove recursion formulas which are much shorter. Many examples are given which compare the different recursion formulas given. Also, in Section 7, we prove Theorem 1.4, and give some comments as to why one could expect the length of the recursion formulas to shorten even further. In Appendix we give several examples with $m = 5, 6, 7, 9, 10, 14$ in their entirety.

2. Resultants, linear recurrences and circulant tori

2.1. **The resultant.** Let $f(t) = \sum_{i=0}^{m} f_i t^i$ and $g(t) = \sum_{j=0}^{n} g_j t^j$ be non-zero polynomials over a commutative ring of degrees m, n respectively. We assume $f_m \neq 0$ and $g_n \neq 0$. The Sylvester matrix $\mathrm{Syl}(f,g)$ is the $(m+n)\times(m+n)$ matrix whose first n rows are successive shifts of the coefficient vector of f and whose last m rows are successive shifts of the coefficient vector of g:

$$\mathrm{Syl}(f,g) = \begin{pmatrix} f_m & f_{m-1} & \cdots & f_0 & 0 & \cdots & 0 \\ 0 & f_m & f_{m-1} & \cdots & f_0 & \cdots & 0 \\ \vdots & & \ddots & \ddots & & \ddots & \vdots \\ 0 & \cdots & 0 & f_m & f_{m-1} & \cdots & f_0 \\ g_n & g_{n-1} & \cdots & g_0 & 0 & \cdots & 0 \\ 0 & g_n & g_{n-1} & \cdots & g_0 & \cdots & 0 \\ \vdots & & \ddots & \ddots & & \ddots & \vdots \\ 0 & \cdots & 0 & g_n & g_{n-1} & \cdots & g_0 \end{pmatrix} \tag{13}$$

The resultant of f and g, denoted by $\mathrm{Res}(f,g)$, is the determinant of $\mathrm{Syl}(f,g)$, written as

$$\mathrm{Res}(f,g) := \det \mathrm{Syl}(f,g).$$

The resultant $\mathrm{Res}(f,g)$ is a polynomial with integer coefficients in the coefficients of f and g. The following lemma summarizes well-known properties of $\mathrm{Res}(f,g)$, which we present here for the convenience of the reader.

Lemma 2.1. *Assume the notation as above and that the commutative ring is a field.*

(i) We have that $\mathrm{Res}(f,g) = 0$ *if and only if positive degree polynomials* f *and* g *have a non-constant common factor.*

(ii) If f *factors completely into linear factors, perhaps in some splitting field, and has roots* $\alpha_1, \ldots, \alpha_m$*, counted with multiplicity, then*

$$\mathrm{Res}(f,g) = f_m^n \prod_{i=1}^{m} g(\alpha_i). \tag{14}$$

In particular, if f *is monic, then* $\mathrm{Res}(f,g) = \prod_{i=1}^{m} g(\alpha_i)$*. Symmetrically, if* g *factors completely into linear factors and has roots* $\beta_1, \ldots, \beta_n$*, then* $\mathrm{Res}(f,g) = (-1)^{mn} g_n^m \prod_{j=1}^{n} f(\beta_j)$*.*

Classically, resultants have been used to show, for example, that the sum of two algebraic integers is an algebraic integer. To do so, one forms the polynomial in y obtained by computing the resultant in x of $\tilde{f}(x) = f(x)$ and $\tilde{g}(x) = g(y-x)$, which we write as $\mathrm{Res}_x(\tilde{f},\tilde{g})(y)$. By (14), $\mathrm{Res}_x(\tilde{f},\tilde{g})(y)$ is zero if and only if $y = \alpha + \beta$ where α is a root of f and β is a root of g. If both $\tilde{f}$ and $\tilde{g}$ are monic, then so is $\mathrm{Res}_x(\tilde{f},\tilde{g})(y)$, thus showing that $\alpha+\beta$ is an algebraic integer if both α and β are algebraic integers.

We will use a similar consideration in this article in our proof of Theorem 1.2.

2.2. **Closed form solutions of linear recurrences.** Assume the sequence $\{a_k\}$ for $k \geq 0$ satisfies an n step linear homogeneous recurrence relation with constant coefficients. In other words, for $k \geq n$ we have that

$$a_k = c_1 a_{k-1} + c_2 a_{k-2} + \cdots + c_n a_{k-n}, \tag{15}$$

with n prescribed initial terms $\{a_0, a_1, \ldots, a_{n-1}\}$. Substituting $a_h = r^h$ for all h into (15) yields the degree n characteristic equation

$$f(r) := r^n - c_1 r^{n-1} - c_2 r^{n-2} - \cdots - c_{n-1} r - c_n = 0. \tag{16}$$

Let $\{r_j\}_{j=1}^n$ denote the roots of the characteristic polynomial (16), computed in a splitting field and counted with multiplicity. If all roots $\{r_1, r_2, \ldots, r_n\}$ are distinct, then the explicit closed form for a_k is given as

$$a_k = C_1(r_1)^k + C_2(r_2)^k + \cdots + C_n(r_n)^k \tag{17}$$

where the constants $C = [C_1, \ldots, C_n]^T$ are obtained by solving the $n \times n$ Vandermonde linear system

$$\begin{pmatrix} 1 & 1 & \cdots & 1 \\ r_1 & r_2 & \cdots & r_n \\ \vdots & \vdots & & \vdots \\ r_1^{n-1} & r_2^{n-1} & \cdots & r_n^{n-1} \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \\ \vdots \\ C_n \end{pmatrix} = \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_{n-1} \end{pmatrix}.$$

If a root r_j repeats with multiplicity μ, its contribution to the solution of a_k is not $C_j r_j^k$ but rather is a polynomial in k of degree $\mu - 1$, namely

$$\big(B_0 + B_1 k + B_2 k^2 + \cdots + B_{\mu-1} k^{\mu-1}\big)(r_j)^k. \tag{18}$$

One no longer uses the Vandermonde linear system to determine the coefficients $\{B_j, C_j\}$, but instead forms the obvious system of equations which determines the coefficients from the first n terms in the recursive sequence.

Define the ordinary generating function of the sequence $\{a_k\}$ as the formal infinite power series $A(x) = \sum_{k=0}^{\infty} a_k x^k$. It can be shown that A is a rational function, meaning for $|x|$ sufficiently small, we have that

$$A(x) = \frac{P(x)}{1 - c_1 x - c_2 x^2 - \cdots - c_n x^n}. \tag{19}$$

The denominator is the reciprocal characteristic polynomial, meaning $r^n f(1/r)$, and the numerator $P(x)$ is a polynomial of degree $\leq n-1$ encoding the initial conditions of the recursive sequence. More precisely, $P(x)$ is given by

$$P(x) = \sum_{i=0}^{n-1} \Big(a_i - \sum_{j=1}^{i} c_j a_{i-j} \Big) x^i.$$

If we factor the denominator into linear terms, then we can expand the denominator using a geometric series for each term and obtain the closed formula (17) for each a_k as sum of k-th powers of the roots of the characteristic polynomial.

For the questions we consider in this article, the characteristic polynomial has distinct roots, so (18) does not play a role.

2.3. **Discrete and circulant tori.** For an integer $m \geq 1$, the discrete circle $\mathbb{Z}/m\mathbb{Z}$ is the graph on m vertices with edges joining nearest neighbors, a 2-regular graph. For $M = (m_1, \ldots, m_k)$, the discrete torus $\mathrm{DT}(M)$ is the Cayley graph of $\prod_j \mathbb{Z}/m_j\mathbb{Z}$ with respect to the standard basis vectors. Throughout, we take $k = 3$, $m_1 = m_2 = m_3 = m$, and write $\mathrm{DT}(m)$ for the resulting 6-regular graph on m^3 vertices.

Set $\mathrm{DT}(m) = (\mathbb{Z}/m\mathbb{Z})^3$. The vertex (x_1, x_2, x_3) is adjacent to (x_1', x_2', x_3') if and only if the two ordered triples differ by ± 1 in exactly one coordinate. Define the quotient graph

$$G_m := \mathrm{DT}(m)/\sim \quad \text{where} \quad (x_1, x_2, x_3) \sim (x_1 + 1, x_2 + 1, x_3 + 1).$$

The cyclic group $Z/m\mathbb{Z}$, which is generated by the diagonal shift which defines G_m, acts freely on $\mathrm{DT}(m)$ with quotient G_m. Let us write (x_1, x_2) for the coset of $(x_1, x_2, 0)$, so that the vertex $(x_1, x_2, x_3) \in \mathrm{DT}(m)$ lies in the coset $(x_1 - x_3, x_2 - x_3)$. By tracking the edges to $(x_1, x_2, 0)$ under the quotient map, we can show that (x_1, x_2) has six edges in G_m. First, we have the four edges to $(x_1 \pm 1, x_2, 0)$ and $(x_1, x_2 \pm 1, 0)$ in $\mathrm{DT}(m)$ which descend to the edges $(x_1, x_2) \sim (x_1 \pm 1, x_2)$ and $(x_1, x_2) \sim (x_1, x_2 \pm 1)$ in G_m. Second, we have the two edges to $(x_1, x_2, \pm 1)$ in $\mathrm{DT}(m)$ which descend to edges $(x_1, x_2) \sim (x_1 \mp 1, x_2 \mp 1)$, respectively, in G_m. Thus G_m is a 6-regular graph

on m^2 vertices with $3m^2$ edges, and $\mathrm{DT}(m) \to G_m$ is an m-to-1 regular covering. We call G_m a *circulant torus*, see Fig. 1.

2.4. **Eigendata on the discrete torus.** For $d \in G_m$, let T_d denote the operator

$$(T_d f)(x) := f(x+d).$$

These operators commute and are unitary, $T_d^* = T_{-d}$.

With this notation one can define a directional Laplacian as follows:

$$\Delta_d = (T_d - I)^*(T_d - I) = 2I - T_d - T_{-d}.$$

Define a corresponding higher-order Laplace operator:

$$\Delta_m = \frac{1}{4}\Delta_{e_1}\Delta_{e_2}\Delta_{e_1+e_2}.$$

This is related to Weyl's denominator formula for SU(3).

An alternative to Δ_m is the following. Note the elementary identity for any real α and β,

$$4\sin\alpha\sin\beta\sin(\alpha+\beta) = \sin 2\alpha + \sin 2\beta - \sin 2(\alpha+\beta).$$

This has the operator manifestation as

$$A_m = \tfrac{i}{6}(T_{e_1} - T_{-e_1} + T_{e_2} - T_{-e_2} + T_{-e_1-e_2} - T_{e_1+e_2}).$$

The triangles can be oriented so that e_1, e_2, and $-e_1-e_2$ provide the corresponding edge orientation. The above operator corresponds to the oriented graph adjacency operator. Note that

$$\Delta_m = 9A_m^* A_m.$$

For $j = (a,b) \in (\mathbb{Z}/m\mathbb{Z})^2$, define

$$\chi_j(x_1,x_2) := \frac{1}{m}\exp\left(2\pi i(ax_1+bx_2)/m\right).$$

These characters form an orthonormal basis of $\ell^2(G_m)$. They also serve as eigenfunctions to the basic operators

$$T_{(u,v)}\chi_j = \exp\left(2\pi i(au+bv)/m\right)\chi_j$$

and hence also of Δ_m.

The spectral zeta function is a sum over the non-zero eigenvalues, more precisely

$$\zeta_{\Delta_m}(s) = \sum_{j \in G_m \setminus Z} \lambda_m(j)^{-s}$$

where $Z = \{j = (a,b) : a = 0, b = 0, \text{or } a+b \equiv 0\}$. The number of terms in this sum is

$$M = m^2 - (3m-2) = (m-1)(m-2).$$

The reduced spectral polynomial is

$$H(w) = \prod_{j \in G_m \setminus Z} (\lambda_m(j) - w)$$

Take its logarithmic derivative and define

$$F(w) = -\frac{1}{m^2}\frac{H'(w)}{H(w)}.$$

This is clearly a rational function.

Let

$$r_{a,b} = -\sin(2\pi a/m) - \sin(2\pi b/m) + \sin(2(a+b)\pi/m).$$

In view of the trigonometric identity just mentioned,

$$r_{a,b}^2 = \lambda_m((a,b)) = 16\sin^2\frac{\pi a}{m}\sin^2\frac{\pi b}{m}\sin^2\frac{\pi(a+b)}{m}.$$

We also define

$$m^2G(w) = m^2G(x,x;w) := \sum_{j\in G_m\setminus Z} \frac{1}{w-r_j},$$

which can be viewed as a resolvent kernel for any $x \in G_m$.

Define another spectral determinant polynomial by

$$D(w) := \prod_{j\in G_m\setminus Z} (w-r_j). \tag{20}$$

Then $D(w)$ is a monic polynomial of degree $(m-1)(m-2)$ in w and whose roots counted with multiplicity are r_j. The involution $j \mapsto -j$ sends r_j to $-r_j$, so D is even. Therefore it can be written $D(w) = \hat{D}(w^2)$. Then $H(w) = \hat{D}(w)^2$.

Note that

$$m^2G(w) = \frac{D'(w)}{D(w)} = \frac{d}{dw}\log D(w)$$

for any vertex $x \in G_m$. Moreover,

$$F(w) := F(x,x;w) = -\frac{1}{2\sqrt{w}}\big(G(x,x;\sqrt{w}) - G(x,x;-\sqrt{w})\big).$$

We will use the following notation. Let us set

$$\Lambda_m(j) := \frac{1}{3}\sum_{\ell=1}^{3} \cos\frac{2\pi(j_\ell + m/4)}{m} \tag{21}$$

Then we have that

$$3\Lambda_m(j) = (-1)^\ell 4\sin\frac{\pi j_1}{m}\sin\frac{\pi j_2}{m}\sin\frac{\pi j_3}{m} \quad \text{whenever } j_1+j_2+j_3 = \ell m. \tag{22}$$

For $(j_1,j_2,j_3) \in (\mathbb{Z}/m\mathbb{Z}\setminus\{0\})^3$ admissible values of ℓ for which $j_1+j_2+j_3 = \ell m$ are $\ell \in \{1,2\}$. Let

$$A(m) := \{j \in (\mathbb{Z}/m\mathbb{Z})^3 : j_1+j_2+j_3 \equiv 0\} \quad \text{and} \quad A_0(m) := \{j \in A(m) : \Lambda_m(j) = 0\}.$$

There are m^3 ordered triples (j,ℓ,k) mod m, of which m^2 are such that $j+\ell+k \equiv 0 \bmod m$, so $A(m)$ has m^2 elements. Among the m^2 ordered triples (j,ℓ,k) mod m with $j+\ell+k \equiv 0$, there are $3m-2$ triples that have at least one coordinate congruent to zero modulo m, so $A_0(m)$ has $3m-2$ elements.

Define $i_m : G_m \to A(m)$ via $i_m(a,b) = (a,b,-a-b)$. Note that $r_{a,b} = 3\Lambda_m(i_m(a,b))$ and $\lambda_m((a,b)) = r_{a,b}^2$.

3. Proof and examples of Theorem 1.1

3.1. Proof of Theorem 1.1.

Proof. Fix $m \geq 3$ and set $c_m = 3m^2/4$. For $j = (a,b) \in G_m$ and $d = (u,v) \in G_m$, the Fourier characters defined above satisfy

$$T_d\chi_j = \exp\left(\frac{2\pi i(au+bv)}{m}\right)\chi_j.$$

Consequently,

$$\begin{aligned}\Delta_d\chi_j &= \left(2 - \exp\left(\frac{2\pi i(au+bv)}{m}\right) - \exp\left(-\frac{2\pi i(au+bv)}{m}\right)\right)\chi_j \\ &= 4\sin^2\left(\frac{\pi(au+bv)}{m}\right)\chi_j.\end{aligned}$$

Since

$$\Delta_m = \frac{1}{4}\Delta_{e_1}\Delta_{e_2}\Delta_{e_1+e_2},$$

we obtain

$$\Delta_m\chi_j = \lambda_m(j)\chi_j, \qquad \lambda_m(j) = 16\sin^2\left(\frac{\pi a}{m}\right)\sin^2\left(\frac{\pi b}{m}\right)\sin^2\left(\frac{\pi(a+b)}{m}\right).$$

The characters form an orthonormal basis of $\ell^2(G_m)$, so these are all the eigenvalues, counted with multiplicity. Moreover, $\lambda_m(j) = 0$ precisely when $a = 0$, $b = 0$, or $a + b = 0$ in $\mathbb{Z}/m\mathbb{Z}$. Thus the nonzero eigenvalues are indexed by $G_m \setminus Z$, and they are strictly positive.

The map

$$(a, b) \longmapsto (a, b, -a - b)$$

is a bijection from $G_m \setminus Z$ onto the set of triples occurring in the definition of $V_n(m)$. For the corresponding triple (a, b, k), we have $k \equiv -a - b \pmod m$, and hence

$$\sin^2\left(\frac{\pi k}{m}\right) = \sin^2\left(\frac{\pi(a+b)}{m}\right).$$

It follows that

$$\left(8\sin\left(\frac{\pi a}{m}\right)\sin\left(\frac{\pi b}{m}\right)\sin\left(\frac{\pi k}{m}\right)\right)^2 = 4\lambda_m((a, b)).$$

Therefore, for every integer $n \geq 1$,

$$\begin{aligned} V_n(m) &= 3^n m^{2n} \sum_{j\in G_m\setminus Z} \big(4\lambda_m(j)\big)^{-n} \\ &= \left(\frac{3m^2}{4}\right)^n \sum_{j\in G_m\setminus Z} \lambda_m(j)^{-n} \\ &= c_m^n \zeta_{\Delta_m}(n). \end{aligned}$$

This proves the first identity.

Next, logarithmic differentiation of the finite product

$$H(w) = \prod_{j\in G_m\setminus Z} \big(\lambda_m(j) - w\big)$$

gives

$$\frac{H'(w)}{H(w)} = -\sum_{j\in G_m\setminus Z} \frac{1}{\lambda_m(j) - w}.$$

By the definition of F, we therefore have

$$m^2F(w) = -\frac{H'(w)}{H(w)} = \sum_{j\in G_m\setminus Z} \frac{1}{\lambda_m(j) - w}.$$

In particular, F is rational and is holomorphic near $w = 0$, since $H(0) \neq 0$.

Set

$$\lambda_* := \min_{j\in G_m\setminus Z} \lambda_m(j) > 0.$$

For $|w| < \lambda_*$, each summand has the geometric expansion

$$\frac{1}{\lambda_m(j) - w} = \sum_{k=0}^{\infty} \frac{w^k}{\lambda_m(j)^{k+1}}.$$

These expansions converge absolutely and uniformly on every closed disk $|w| \le \rho < \lambda_*$. Since the sum over j is finite, we may interchange the summations to obtain

$$\begin{aligned} m^2 F(w) &= \sum_{k=0}^{\infty} w^k \sum_{j \in G_m \setminus Z} \lambda_m(j)^{-k-1} \\ &= \sum_{n=1}^{\infty} \zeta_{\Delta_m}(n) w^{n-1} \\ &= \sum_{n=1}^{\infty} c_m^{-n} V_n(m) w^{n-1}. \end{aligned}$$

Taking the $(n-1)$-st derivative at $w = 0$ yields

$$m^2 F^{(n-1)}(0) = (n-1)!\, c_m^{-n} V_n(m).$$

Since $c_m^n = 3^n (m/2)^{2n}$, this is equivalent to

$$V_n(m) = 3^n \left(\frac{m}{2}\right)^{2n} \frac{m^2}{(n-1)!} \frac{\partial^{n-1}}{\partial w^{n-1}} F(w)\Big|_{w=0},$$

as claimed.

Finally, multiplying the power-series expansion of $m^2 F(w)$ by w and substituting $w = c_m \tau$, we obtain

$$c_m \tau\, m^2 F(c_m \tau) = \sum_{n=1}^{\infty} c_m^{-n} V_n(m) (c_m \tau)^n = \sum_{n=1}^{\infty} V_n(m) \tau^n.$$

This identity holds analytically for

$$|\tau| < \frac{\lambda_*}{c_m},$$

and the series converges absolutely and uniformly on every closed disk $|\tau| \le R < \lambda_*/c_m$.

The same geometric expansions are valid as formal power series, because every factor $\lambda_m(j) - w$ has a nonzero constant term. Hence the generating-function identity also holds as an identity of formal power series in τ. □

3.2. **Examples.** The examples in this section are built using the precise information for the spectrum of G_m. Admittedly, these computations are somewhat inefficient. We are presenting them here as a benchmark for comparison with later examples.

Example 3.1 ($m = 3$). The two nonzero eigenvalues are $r = \pm\frac{3\sqrt{3}}{2}$, each of multiplicity 1. Then

$$m^2 G(w) = \frac{1}{w - \frac{3\sqrt{3}}{2}} + \frac{1}{w + \frac{3\sqrt{3}}{2}} = \frac{2w}{w^2 - \frac{27}{4}} = \frac{8w}{4w^2 - 27}.$$

Since $m^2 = 9$, we get that

$$G(w) = \frac{8w}{9(4w^2 - 27)}$$

and

$$F(w) = -\frac{G(\sqrt{w})}{\sqrt{w}} = -\frac{8}{9(4w - 27)} = \frac{8}{9(27 - 4w)}.$$

When expanding as a geometric series about $w = 0$, we have that

$$F(w) = \frac{8}{243} \sum_{k \ge 0} \Big(\frac{4w}{27}\Big)^k = \sum_{k \ge 0} \frac{8}{243} \Big(\frac{4}{27}\Big)^k w^k,$$

so $\partial_w^{n-1}F(w)\big|_{w=0} = (n-1)!\,\frac{8}{243}\big(\frac{4}{27}\big)^{n-1}$, and

$$V_n(3) = 3^n\Big(\frac{3}{2}\Big)^{2n}\cdot 9\cdot\frac{8}{243}\Big(\frac{4}{27}\Big)^{n-1} = 2$$

for every $n\geq 1$. Thus, $V_n(3)=2$ for all $n\geq 1$. Separately, since $c_3 = 3(3/2)^2 = 27/4$, we get that

$$\sum_{n\geq 1} V_n(3)\,\tau^n = c_3\tau\cdot 9\cdot F(x,x;c_3\tau) = \frac{27}{4}\tau\cdot 9\cdot\frac{8}{9\big(27-27\tau\big)} = \frac{2\tau}{1-\tau} = 2\sum_{n\geq 1}\tau^n,$$

again giving $V_n(3)=2$ for every $n\geq 1$.

Example 3.2 ($m=5$). The twelve nonzero eigenvalues fall into four classes of values each with multiplicity 3. Specifically, the eigenvalues are of the form $r_j = \pm a_1$ and $r_j = \pm a_2$ where

$$a_1 = \sqrt{\frac{25+5\sqrt{5}}{8}}, \qquad a_2 = \sqrt{\frac{25-5\sqrt{5}}{8}}.$$

Then

$$m^2G(w) = \frac{3\cdot 2w}{w^2-a_1^2} + \frac{3\cdot 2w}{w^2-a_2^2} = \frac{6w}{w^2-\frac{25+5\sqrt{5}}{8}} + \frac{6w}{w^2-\frac{25-5\sqrt{5}}{8}}.$$

When combining over a common denominator, using $a_1^2+a_2^2 = \frac{25}{4}$, $a_1^2a_2^2 = \frac{125}{16}$, and $m^2=25$ gives

$$G(w) = \frac{192w^3-600w}{400w^4-2500w^2+3125}$$

and then

$$F(w) = -\frac{G(\sqrt{w})}{\sqrt{w}} = \frac{600-192w}{400w^2-2500w+3125}.$$

When expanding about $w=0$, we get that

$$V_1(5)=90, \qquad V_2(5)=810, \qquad V_3(5)=8100, \qquad V_4(5)=85050.$$

With $c_5 = 3\cdot 25/4 = 75/4$,

$$\sum_{n\geq 1} V_n(5)\,\tau^n = c_5\tau\cdot 25\cdot F(c_5\tau) = \frac{90\tau(1-6\tau)}{45\tau^2-15\tau+1}.$$

Now when expanding in τ reproduces $V_1(5),\ldots,V_4(5)$ above, and the two poles of the denominator $45\tau^2-15\tau+1=0$ are $\tau = a_1^2/c_5$ and $\tau = a_2^2/c_5$.

Example 3.3 ($m=6$). By direct evaluation, we have that $\Lambda_6(j) = \frac{1}{3}\sum_{\ell=1}^{3}\cos\big(2\pi(j_\ell+\frac{3}{2})/6\big)$ over the $20=(6-1)(6-2)$ triples of $A(6)\setminus A_0(6)$ produces exactly three classes, namely

$$r = \pm\frac{3\sqrt{3}}{2}\ \text{(mult. 1)}, \qquad r=\pm\sqrt{3}\ \text{(mult. 6)}, \qquad r=\pm\frac{\sqrt{3}}{2}\ \text{(mult. 3)},$$

with total multiplicity being $1+1+6+6+3+3=20=(6-1)(6-2)$ as required. Then

$$m^2G(w) = \frac{2w}{w^2-\frac{27}{4}} + \frac{12w}{w^2-3} + \frac{6w}{w^2-\frac{3}{4}},$$

which, combined over a common denominator, using $m^2=36$ becomes

$$G(w) = \frac{80w^5-624w^3+747w}{144w^6-1512w^4+3969w^2-2187}$$

and then

$$F(w) = -\frac{G(x,x;\sqrt{w})}{\sqrt{w}} = \frac{-80w^2+624w-747}{144w^3-1512w^2+3969w-2187}.$$

Expanding $F(w)$ about $w=0$ and applying $V_n(m) = 3^n(m/2)^{2n} \cdot m^2/(n-1)! \cdot \partial_w^{n-1} F(x,x;w)\big|_{w=0}$ term by term gives

$$V_1(6) = 332, \quad V_2(6) = 8780, \quad V_3(6) = 288812, \quad V_4(6) = 10156940.$$

With $c_6 = 3 \cdot 6^2/4 = 27$,

$$\sum_{n\geq 1} V_n(6)\,\tau^n = c_6\tau \cdot 36 \cdot F(x,x;c_6\tau) = \frac{4\tau\big(6480\tau^2 - 1872\tau + 83\big)}{(1-4\tau)(1-9\tau)(1-36\tau)}.$$

By expanding in τ reproduces $V_1(6), \ldots, V_4(6)$ as above.

Example 3.4 ($m = 7$). The thirty nonzero eigenvalues fall into four classes. One with the pair $r = \pm\sqrt{7}/2$, of multiplicity 6, and then together with three pairs $r = \pm\sqrt{x_k}$ ($k = 0, 1, 2$), each of multiplicity 3, where x_0, x_1, x_2 are the three real roots of

$$Q(x) := 64x^3 - 560x^2 + 1176x - 343.$$

Rather than substitute the individual radical values of the x_k, we use the elementary identity

$$\sum_{k=0}^{2} \frac{1}{w^2 - x_k} = \frac{Q'(w^2)}{Q(w^2)},$$

which involves no radicals at all. Then

$$m^2 G(w) = \frac{12w}{w^2 - \frac{7}{4}} + 6w \cdot \frac{Q'(w^2)}{Q(w^2)} = \frac{7680w^7 - 61824w^5 + 131712w^3 - 65856w}{256w^8 - 2688w^6 + 8624w^4 - 9604w^2 + 2401},$$

with $m^2 = 49$,

$$G(w) = \frac{7680w^7 - 61824w^5 + 131712w^3 - 65856w}{12544w^8 - 131712w^6 + 422576w^4 - 470596w^2 + 117649}$$

then

$$F(w) = -\frac{G(x,x;\sqrt{w})}{\sqrt{w}} = \frac{-7680w^3 + 61824w^2 - 131712w + 65856}{12544w^4 - 131712w^3 + 422576w^2 - 470596w + 117649}.$$

Expanding $F(x,x;w)$ about $w=0$ and applying $V_n(7) = 3^n \cdot (7/2)^{2n} \cdot 49/(n-1)! \cdot \partial_w^{n-1} F(w)\big|_{w=0}$ term by term gives

$$V_1(7) = 1008, \qquad V_2(7) = 74088, \qquad V_3(7) = 7279146.$$

With $c_7 = 3 \cdot 7^2/4 = 147/4$,

$$\sum_{n\geq 1} V_n(7)\,\tau^n = c_7\tau \cdot 49 \cdot F(c_7\tau),$$

which, expanded in τ, reproduces $V_1(7), V_2(7), V_3(7)$ above.

4. A resultant using Chebyshev polynomials

We now give our first construction of $D(w)$ as a resultant, or rather as a factor of a resultant, which is built directly from Chebyshev-type polynomials. For $j = 0, \ldots, m-1$ let

$$x_j := -\sin(2\pi j/m),$$

the real numbers whose sums generate r_j. Recall the classical Chebyshev identity

$$T_m\Big(\tfrac{1}{2}(u + u^{-1})\Big) = \tfrac{1}{2}(u^m + u^{-m}) \quad \text{for any } u \neq 0 \text{ and } m \geq 0.$$

Lemma 4.1. *Let $\zeta := e^{2\pi i/m}$ and, for $j = 0, \dots, m-1$. Set $u_j := i\zeta^j$, from which we have that $x_j = \mathrm{Re}(u_j)$. Then*

$$T_m(x_j) = \mathrm{Re}(i^m) = \begin{cases} 0, & m \text{ odd}, \\ (-1)^{m/2}, & m \text{ even}. \end{cases}$$

Proof. Since $|u_j| = 1$, we have $u_j^{-1} = \overline{u_j}$, so $x_j = \frac{1}{2}(u_j + u_j^{-1})$, and the identity above gives

$$T_m(x_j) = \tfrac{1}{2}\big(u_j^m + u_j^{-m}\big) = \tfrac{1}{2}\big(i^m\zeta^{jm} + i^{-m}\zeta^{-jm}\big) = \tfrac{1}{2}\big(i^m + i^{-m}\big) = \mathrm{Re}(i^m),$$

from which the lemma follows. □

Define

$$Q_m(x) := \begin{cases} T_m(x), & m \text{ odd}, \\ T_m(x) - i^m, & m \text{ even}, \end{cases}$$

which is a polynomial with integer coefficients, of degree m and with leading coefficient $c := 2^{m-1}$ for both odd and even m. By the Lemma 4.1, the m values $x_0, \dots, x_{m-1}$ are roots of Q_m. Since $\deg Q_m = m$, these roots, counted with multiplicity, are all of the roots of Q_m.

Define

$$R(w) := \prod_{j=0}^{m-1}\prod_{k=0}^{m-1} Q_m(w - x_j - x_k).$$

Theorem 4.2. *For every $m \geq 3$, $R(w)$ is expressible as an iterated resultant, namely that*

$$R(w) = c^{-2m^2}\,\mathrm{Res}_{x_1}\Big(Q_m(x_1),\ \mathrm{Res}_{x_2}\big(Q_m(x_2),\, Q_m(w - x_1 - x_2)\big)\Big).$$

Furthermore, $R(w)$ factors as

$$R(w) = D(w)\cdot w^{3m-2}\cdot \Phi(w),$$

where $D(w)$ is degree $(m-1)(m-2)$ polynomial defined by (20) and Φ is another polynomial.

Remark 4.3. The factor w^{3m-2} accounts for the $3m-2$ triples in $A(m)$ with at least one zero component, and $\Phi(w)$ is an explicit polynomial of degree $m^2(m-1)$, coming from index pairs (j,k) whose associated third index j_3 which does not yield a triple summing to 0 mod m.

Proof. Since $Q_m(x_2)$ has degree m and leading coefficient c, the product formula for resultants gives $\mathrm{Res}_{x_2}(Q_m(x_2), Q_m(w - x_1 - x_2)) = c^m \prod_{k=0}^{m-1} Q_m(w - x_1 - x_k)$, a polynomial in x_1 of degree m^2. Applying the resultant formula once more, now for $Q_m(x_1)$, we get that

$$\begin{aligned}\mathrm{Res}_{x_1}\Big(Q_m(x_1),\, c^m\prod_k Q_m(w - x_1 - x_k)\Big) &= c^{m^2}\prod_{j=0}^{m-1}\Big[c^m\prod_{k=0}^{m-1} Q_m(w - x_j - x_k)\Big] \\ &= c^{2m^2}\prod_{j,k} Q_m(w - x_j - x_k) = c^{2m^2}R(w),\end{aligned}$$

giving the stated formula after dividing by c^{2m^2}.

From the properties of resultants, $Q_m(w - x_j - x_k) = 0$ if and only if $w - x_j - x_k = x_\ell$ for some $\ell \in \{0, \dots, m-1\}$. By Lemma 4.1, the roots of Q_m are exactly the x_ℓ. Therefore,

$$\begin{aligned} R(w) &= c^{m^2} \prod_{j=(j_1,j_2,j_3)\in\{0,\dots,m-1\}^3} (w - x_{j_1} - x_{j_2} - x_{j_3}) \\ &= \prod_{j\in A(m)\setminus A_0(m)} (w - x_{j_1} - x_{j_2} - x_{j_3}) \prod_{j\in A_0(m)} (w - x_{j_1} - x_{j_2} - x_{j_3}) \cdot \\ &\cdot c^{m^2} \prod_{j\in\{0,\dots,m-1\}^3\setminus A(m)} (w - x_{j_1} - x_{j_2} - x_{j_3}) \\ &= D(w) \cdot w^{3m-2} \cdot \Phi(w). \end{aligned}$$

□

Remark 4.4. Theorem 4.2. expresses $D(w)$, exactly, as a factor of an explicit resultant, with an explicit overall constant and a complementary factor $F(w)$. What it does not give us is a way to extract $D(w)$ alone without already knowing enough about the index set $A(m)$ to separate the two factors. This is exactly the sense in which the resultant is "too big". It ranges, unavoidably, over the full m^2 pairs of roots of Q_m with no built-in mechanism for restricting to the constrained triples defining $A(m)$. Below we develop a second construction that resolves this issue, by working with another variable that does encode the constraint directly, and, unlike the present construction, requires no case distinction on the parity of m at all.

Example 4.5 ($m = 3$). From Example 3.1 we know that $D(w) = \dfrac{4w^2 - 27}{4}$. Indeed, when computing $R(w)$, which has degree 27 and dividing by $w^7 D(w)$ leaves zero remainder and quotient

$$\Phi(w) = 2^6\,(w^2 - 3)^3 (4w^2 - 3)^6,$$

of degree $18 = 3^2(3 - 1)$ exactly as predicted.

Example 4.6 ($m = 5$). From Example 3.2 $D(w) = \dfrac{(16w^4 - 100w^2 + 125)^3}{4096}$. In this case, $R(w)$ has degree 125. When dividing by $w^{13} D(w)$ we get the quotient

$$\Phi(w) = 2^{12}\big(w^4 - 5w^2 + 5\big)^3\big(16w^4 - 180w^2 + 405\big)\big(16w^4 - 100w^2 + 5\big)^3$$

$$\times\ \big(16w^4 - 40w^2 + 5\big)^6\big(16w^4 - 20w^2 + 5\big)^{12},$$

which has degree $100 = 5^2(5 - 1)$.

Example 4.7 ($m = 6$). Since m is even, $Q_6(x) = T_6(x) + 1 = 32x^6 - 48x^4 + 18x^2 + 1$. From Example 3.3 $D(w) = \big(w^2 - \frac{27}{4}\big)(w^2 - 3)^6\big(w^2 - \frac{3}{4}\big)^3$. The polynomial $R(w)$ has degree 216, and its quotient by $w^{16} D(w)$ is

$$\Phi(w) = 2^{76}\, w^{40} (w^2 - 3)^{18} (4w^2 - 27)^7 (4w^2 - 3)^{45},$$

which has degree $180 = 6^2(6 - 1)$.

Example 4.8 ($m = 7$). From Example 3.4,

$$D(w) = \frac{(4w^2 - 7)^6 (64w^6 - 560w^4 + 1176w^2 - 343)^3}{2^{30}},$$

of degree 30. The polynomial $R(w)$ has degree 343, and the quotient by $w^{19}D(w)$ is

$$\begin{aligned}\Phi(w) = 2^{18}&\big(w^6-7w^4+14w^2-7\big)^3\big(64w^6-1008w^4+4536w^2-5103\big)\\ &\times\big(64w^6-560w^4+616w^2-7\big)^3\big(64w^6-560w^4+952w^2-7\big)^3\big(64w^6-560w^4+1512w^2-1183\big)^3\\ &\times\big(64w^6-336w^4+140w^2-7\big)^6\big(64w^6-224w^4+84w^2-7\big)^6\big(64w^6-224w^4+196w^2-7\big)^6\\ &\times\big(64w^6-112w^4+56w^2-7\big)^{18},\end{aligned}$$

which has degree $294 = 7^2(7-1)$.

5. An efficient resultant construction

We now give a refined resultant construction that avoids the factor $\Phi(w)$, instead yielding the function $D(w)$, w^{3m-2}, and a multiplicative constant.

Remark 5.1. It is natural to ask whether the construction of Section 4 can be refined by imposing directly the angle-sum constraint on the real variables $x_\ell = \cos\widetilde{\theta}_\ell$, $\widetilde{\theta}_\ell := \frac{2\pi(j_\ell+m/4)}{m}$. For a constrained triple $j_1+j_2+j_3 \equiv 0 \mod m$, we have $\widetilde{\theta}_1+\widetilde{\theta}_2+\widetilde{\theta}_3 \equiv \frac{3\pi}{2} \pmod{2\pi}$. Hence, after writing $s_\ell = \sin\widetilde{\theta}_\ell$, one obtains

$$x_3 = \cos\big(\tfrac{3\pi}{2} - \widetilde{\theta}_1 - \widetilde{\theta}_2\big) = -\sin(\widetilde{\theta}_1+\widetilde{\theta}_2) = -(s_1x_2 + x_1s_2).$$

Eliminating s_1 and s_2 by using $s_\ell^2 = 1-x_\ell^2$ gives the polynomial relation

$$P(x_1,x_2,x_3) := \big(x_3^2 - x_1^2 - x_2^2 + 2x_1^2x_2^2\big)^2 - 4x_1^2x_2^2(1-x_1^2)(1-x_2^2) = 0. \tag{23}$$

Relation (23) is a necessary polynomial relation among the three real parts, but it is not sufficient to recover the original angle-sum constraint. Indeed, the squaring used to eliminate s_1 and s_2 loses their signs. Equivalently, the map $z \mapsto \mathrm{Re}(z)$ on the unit circle identifies z with $\overline{z} = z^{-1}$. Thus, from x_1 and x_2 alone one cannot select the particular phases needed to determine the unique completion $z_{j_3} = -\frac{i}{z_{j_1}z_{j_2}}$, $j_3 \equiv -(j_1+j_2) \pmod m$. Consequently, eliminating x_1, x_2, x_3 using (23) together with the one-variable equations satisfied by the x_j generally introduces extraneous branches and does not isolate the triples in $A(m)$.

This is why, in the next subsection, we pass to the unit-circle variables u and v.

5.1. **The second resultant.** Let $\zeta := e^{2\pi i/m}$. For $j \in \mathbb{Z}/m\mathbb{Z}$, set $z_j := i\zeta^j$.

Lemma 5.2. *The map $j \mapsto z_j$ is a bijection of $\mathbb{Z}/m\mathbb{Z}$ onto the set of roots of $Q(u) := u^m - i^m$. Each z_j lies on the unit circle, and $\mathrm{Re}(z_j) = -\sin(2\pi j/m) = x_j$.*

Proof. Clearly, $(i\zeta^j)^m = i^m\zeta^{jm} = i^m$ since $\zeta^m = 1$. The m values ζ^j are distinct, so the z_j are distinct. Because Q has exactly m roots, the map is onto. By writing $\theta_j := \pi/2 + 2\pi j/m$, we have $z_j = e^{i\theta_j}$, $|z_j| = 1$, and $\mathrm{Re}(z_j) = \cos\theta_j = \cos(\pi/2 + 2\pi j/m) = -\sin(2\pi j/m) = x_j$. □

Lemma 5.3. *For $j_1, j_2, j_3 \in \mathbb{Z}/m\mathbb{Z}$,*

$$j_1+j_2+j_3 \equiv 0 \pmod m \quad \text{if and only if} \quad z_{j_1}z_{j_2}z_{j_3} = -i.$$

Thus, for any two roots $u = z_{j_1}$, $v = z_{j_2}$ of $Q(x) = x^m - i^m$, the value $z_3 := -i/(uv)$ is automatically a root of Q and is equal to z_{j_3}.

Proof. Clearly, $z_{j_1}z_{j_2}z_{j_3} = i^3\zeta^{j_1+j_2+j_3} = -i\cdot\zeta^{j_1+j_2+j_3}$, which equals $-i$ precisely when $\zeta^{j_1+j_2+j_3} = 1$, or equivalently when $j_1+j_2+j_3 \equiv 0 \pmod m$. For the second statement, $uv = -\zeta^{j_1+j_2}$, and $z_3 = -i/(uv) = -i/(-\zeta^{j_1+j_2}) = i\zeta^{-(j_1+j_2)} = z_{-(j_1+j_2) \bmod m} = z_{j_3}$. □

Let us now define the auxiliary polynomial

$$W(u,v,w) := iu^2v^2 + u^2v + uv^2 - 2wuv + u + v - i. \tag{24}$$

Lemma 5.4. *Let u, v lie on the unit circle and set $z_3 := -i/(uv)$. Then*

$$W(u,v,w) = -2uv\Big(w - \big(\mathrm{Re}(u) + \mathrm{Re}(v) + \mathrm{Re}(z_3)\big)\Big).$$

In particular, for $u = z_{j_1}$, $v = z_{j_2}$, $j_3 = -(j_1 + j_2) \bmod m$, we have that

$$W(z_{j_1}, z_{j_2}, w) = -2z_{j_1}z_{j_2}\big(w - 3\Lambda_m(j_1, j_2, j_3)\big).$$

Proof. Since $|u| = |v| = |z_3| = 1$, we have $\bar{u} = 1/u$, and similarly for v and z_3. Then,

$$2\big(\mathrm{Re}(u) + \mathrm{Re}(v) + \mathrm{Re}(z_3)\big) = (u + 1/u) + (v + 1/v) + (z_3 + 1/z_3).$$

By substituting $z_3 = -i/(uv)$, so $1/z_3 = iuv$, and multiplying through by uv, we get that

$$2uv\big(\mathrm{Re}(u) + \mathrm{Re}(v) + \mathrm{Re}(z_3)\big) = u^2v + v + uv^2 + u - i + iu^2v^2 = W(u,v,0).$$

Since the only w-dependence in W is the term $-2wuv$, we get

$$W(u,v,w) = W(u,v,0) - 2wuv = -2uv\Big(w - \big(\mathrm{Re}(u) + \mathrm{Re}(v) + \mathrm{Re}(z_3)\big)\Big).$$

From Lemma 5.2 and Lemma 5.3, we have that

$$\mathrm{Re}(z_{j_1}) + \mathrm{Re}(z_{j_2}) + \mathrm{Re}(z_{j_3}) = x_{j_1} + x_{j_2} + x_{j_3} = 3\Lambda_m(j_1, j_2, j_3),$$

which completes the proof. □

Theorem 5.5. *Let $Q(u) := u^m - i^m$, which is monic of degree m, and W as in (24). Define*

$$\widetilde{D}(w) := \prod_{j_1=0}^{m-1}\prod_{j_2=0}^{m-1} W(z_{j_1}, z_{j_2}, w). \tag{25}$$

Then

$$\widetilde{D}(w) = \mathrm{Res}_u\left(Q(u), \mathrm{Res}_v(Q(v), W(u,v,w))\right)$$

and

$$\widetilde{D}(w) = 2^{m^2} w^{3m-2} D(w),$$

where $D(w)$ is the spectral determinant polynomial defined by (20).

Proof. Let us write $j_3 = j_3(j_1, j_2)$ viewing j_3 as a function of j_1, j_2 when $j_3 := -(j_1 + j_2) \bmod m$. From Lemma 5.4 we have that

$$\begin{aligned}\widetilde{D}(w) &= \prod_{(j_1,j_2)\in\{0,\dots,m-1\}^2}\Big(-2z_{j_1}z_{j_2}(w - 3\Lambda_m(j_1,j_2,j_3))\Big)\\ &= (-2)^{m^2}\Big(\prod_{(j_1,j_2)\in\{0,\dots,m-1\}^2} z_{j_1}z_{j_2}\Big)\Big(\prod_{(j_1,j_2)\in\{0,\dots,m-1\}^2}(w - 3\Lambda_m(j_1,j_2,j_3))\Big).\end{aligned}$$

There are two points to address. First, the reduction of the product from the set of all j_1, j_2, and second the evaluation of the multiplication constant.

Let $P_0 := \prod_{j=0}^{m-1} z_j$ and note that

$$P_0 = \prod_{j=0}^{m-1} z_j = i^m \zeta^{0+1+\cdots+(m-1)} = i^m \zeta^{m(m-1)/2}.$$

Since $\zeta^{m(m-1)/2} = e^{i\pi(m-1)} = (-1)^{m-1}$, we get $P_0 = i^m(-1)^{m-1}$. Then

$$\prod_{(j_1,j_2)\in\{0,\dots,m-1\}^2} z_{j_1}z_{j_2} = P_0^{2m} = i^{2m^2} = (-1)^{m^2}.$$

With this, we get that

$$(-2)^{m^2}\Big(\prod_{(j_1,j_2)\in\{0,\dots,m-1\}^2} z_{j_1}z_{j_2}\Big) = 2^{m^2}.$$

As (j_1, j_2) ranges the distinct m^2 pairs in $(\mathbb{Z}/m\mathbb{Z})^2$, $(j_1, j_2) \mapsto (j_1, j_2, j_3(j_1, j_2))$ is a bijection onto $A(m)$. Every pair (j_1, j_2) produces a unique triple in $A(m)$, by Lemma 5.3. Therefore,

$$\prod_{(j_1,j_2)\in\{0,\dots,m-1\}^2} (w - 3\Lambda_m(j_1, j_2, j_3)) = \prod_{j\in A(m)} (w - 3\Lambda_m(j)) \\ = \prod_{j\in A_0(m)} (w - 3\Lambda(j)) \times \prod_{j\in A(m)\setminus A_0(m)} (w - 3\Lambda_m(j))$$

Each $j \in A_0(m)$ is such that $\Lambda_m(j) = 0$, and $A_0(m)$ has $3m - 2$ elements. Therefore,

$$\prod_{j_1,j_2} (w - 3\Lambda_m(j_1, j_2, j_3)) = w^{3m-2} \prod_{j\in A(m)\setminus A_0(m)} (w - 3\Lambda_m(j)) = w^{3m-2} D(w).$$

□

Remark 5.6. We have completed the task of writing the spectral determinant polynomial as an explicitly computable elementary function. Namely, for every $m \geq 3$ and every w with $w \neq 0$, we have that

$$D(w) = \frac{1}{2^{m^2} w^{3m-2}} \operatorname{Res}_u \Big(Q(u), \operatorname{Res}_v \big(Q(v), W(u, v, w) \big) \Big)$$

where $Q(u) := u^m - i^m$ with W as in (24). From our point of view, this expression is the SU(3) Verlinde sum analogue of the Chebyshev polynomial T_m. Also, because $m \geq 3$, one can group the factors in (25) in conjugate pairs. Therefore, polynomial(25) has real coefficients. Furthermore, the Galois invariance of the root multiset $\{3\Lambda_m(j) : j \in A(m)\}$ when considering the Galois group of the m-th cyclotomic field implies that (25) has rational coefficients.

5.2. **Examples.** Recall that $\deg D = (m-1)(m-2)$, $|A_0(m)| = 3m - 2$, and $\deg \widetilde{D} = (3m-2) + (m-1)(m-2) = m^2$.

Example 5.7 ($m = 3$). $D(w) = \frac{4w^2-27}{4}$, and $\widetilde{D}(w) = 2^9 w^7 D(w) = 512w^7 \cdot \frac{4w^2-27}{4} = 128w^7(4w^2 - 27) = 512w^9 - 3456w^7$.

Example 5.8 ($m = 5$). $D(w) = \frac{(16w^4-100w^2+125)^3}{4096}$, and $\widetilde{D}(w) = 2^{25} w^{13} D(w) = 8192\, w^{13}(16w^4 - 100w^2 + 125)^3$.

Example 5.9 ($m = 6$). $D(w) = \big(w^2 - \frac{27}{4}\big)(w^2 - 3)^6 \big(w^2 - \frac{3}{4}\big)^3$, so

$$\widetilde{D}(w) = 2^{36} w^{16} D(w) = 2^{36}\, w^{16} \Big(w^2 - \tfrac{27}{4}\Big)(w^2 - 3)^6 \Big(w^2 - \tfrac{3}{4}\Big)^3,$$

of degree $36 = m^2$, as required.

Example 5.10 ($m = 7$). $D(w) = \dfrac{(4w^2 - 7)^6 (64w^6 - 560w^4 + 1176w^2 - 343)^3}{2^{30}}$, so

$$\widetilde{D}(w) = 2^{49} w^{19} D(w) = 2^{19}\, w^{19}\, (4w^2 - 7)^6\, (64w^6 - 560w^4 + 1176w^2 - 343)^3,$$

of degree $49 = m^2$, as required.

6. Completing the proof of Theorem 1.2 and recursion formulas

We now complete the proof of Theorem 1.2 by discussing the recursion formulas that follow from the evaluation of the resolvent as a rational function.

6.1. **Proof of the theorem.** Before we begin with the proof, note that the right-hand side of (9) is, indeed, a power series in τ because D is even, so D'/D is odd. Hence, $v \mapsto vD'(v)/D(v)$ an even function of v.

Proof. For notational convenience, set

$$S_{2n}(m) := \sum_{j \in A(m)\setminus A_0(m)} (3\Lambda_m(j))^{-2n}$$

and recall that $D(v) = \prod_{j\in A(m)\setminus A_0(m)}(v - r_j)$, $r_j := 3\Lambda_m(j)$. Then

$$v\frac{D'(v)}{D(v)} = v\sum_j \frac{1}{v-r_j} = \sum_j \left(1 + \frac{r_j}{v-r_j}\right) = (m-1)(m-2) + \sum_j \frac{r_j}{v-r_j},$$

using $\deg D = (m-1)(m-2)$. Hence

$$(m-1)(m-2) - v\frac{D'(v)}{D(v)} = -\sum_j \frac{r_j}{v-r_j} = \sum_j \frac{1}{1-v/r_j} = \sum_j \sum_{k\geq 0} \left(\frac{v}{r_j}\right)^k = \sum_{k\geq 0} v^k \sum_j r_j^{-k},$$

valid as a formal power series in v for $|v|$ smaller than every $|r_j|$. Since, as shown in the previous proof, the eigenvalues r_j occur in $\pm$ pairs, we have $\sum_j r_j^{-k} = 0$ for every odd k, leaving only the even k terms, which are exactly the values $S_{2n}(m)$ at $k = 2n$.

Note that by definition $V_n(m) = 3^n(m/2)^{2n}S_{2n}(m)$, so setting $v := \sqrt{c_m\tau}$ hence $v^{2n} = c_m^n\tau^n = 3^n(m/2)^{2n}\tau^n$ we get $\sum_n V_n(m)\tau^n = (m-1)(m-2) - vD'(v)/D(v)\big|_{v=\sqrt{c_m\tau}}$, as claimed. □

Remark 6.1. Theorem 1.2 is the precise sense in which the generating series is "the log-derivative of the resultant". Again, D is not an unknown polynomial but rather the resultant of two explicit elementary polynomials Q, W in three variables, divided by an explicit monomial.

6.2. **A recursion for the Verlinde Sums for fixed m.** We now show how expressing the generating function as a rational function leads to recurrence formulas.

6.2.1. *From the generating series to a polynomial recursion.* Theorem 1.2 already gives $\sum_n V_n(m)\tau^n$ as an explicit function of D and D'. We now simply record its consequence as a rational function $P(\tau)/Q(\tau)$ of polynomials, from which the recursion coefficients can be read off directly.

Fix $m \geq 3$. Let $D(u)$ be the spectral determinant polynomial, even of degree $M = (m-1)(m-2)$, $D(0) \neq 0$. Let $N(u) := D'(u)$, odd of degree $M-1$. Since D is even and N odd,

$$\widehat{D}(w) := D(u)\big|_{u^2=w}, \qquad \widehat{N}(w) := \frac{N(u)}{u}\bigg|_{u^2=w}$$

are genuine polynomials in w, of degrees $K := M/2$ and $K-1$ respectively.

Corollary 6.2. *Recall the notation $c_m := 3(m/2)^2 = 3m^2/4$. Define*

$$Q(\tau) := \widehat{D}(c_m\tau), \qquad P(\tau) := M\widehat{D}(c_m\tau) - c_m\tau\widehat{N}(c_m\tau),$$

polynomials of degree $K = (m-1)(m-2)/2$ and at most K in τ. Then, with $V_0(m) := M$,

$$\sum_{n\geq 0} V_n(m)\tau^n = \frac{P(\tau)}{Q(\tau)}.$$

Proof. By Theorem 1.2, $\sum_n V_n(m)\tau^n = M - vD'(v)/D(v)\big|_{v=\sqrt{c_m\tau}} = M - vN(v)/D(v)\big|_{v=\sqrt{c_m\tau}}$. Since $N(v)/v$ and $D(v)$ are functions of v^2 (as N odd, D even), $v\cdot N(v)/D(v) = v^2\cdot\widehat{N}(v^2)/\widehat{D}(v^2) = c_m\tau\cdot\widehat{N}(c_m\tau)/\widehat{D}(c_m\tau)$. This gives $\sum_n V_n(m)\tau^n = M - c_m\tau\widehat{N}(c_m\tau)/\widehat{D}(c_m\tau) = P(\tau)/Q(\tau)$. □

Corollary 6.3. *Write* $Q(\tau)=\sum_{k=0}^{K} q_k\tau^k$, $P(\tau)=\sum_{k=0}^{\widetilde{K}} p_k\tau^k$, *where* $\widetilde{K}=\deg P\le K$. *Then*

$$\sum_{k=0}^{\min(n,K)} q_k V_{n-k}(m)=\begin{cases} p_n, & 0\le n\le \widetilde{K},\\ 0, & n>\widetilde{K}.\end{cases} \tag{26}$$

This is a linear recursion of order $K=(m-1)(m-2)/2$ *with constant coefficients, determining* $V_n(m)$ *for* $n>K$ *from* $V_1(m),\dots,V_K(m)$.

Proof. Write $Q(\tau)\sum_n V_n(m)\tau^n=P(\tau)$ and then equate the coefficients of τ^n. □

Remark 6.4. If $\mathrm{GCD}(P(\tau),Q(\tau))$ is a polynomial of degree $d>0$, the linear recursion (26) can, of course, be further reduced to a linear recursion of order $K-d$ with constant coefficients.

6.3. **Recursion formula examples.** Let us record the order $K=(m-1)(m-2)/2$ for the first five levels $m=3,4,5,6,7$. Each recursion was derived symbolically from $D(w)$ via $\widehat{D}(x):=D(w)|_{w^2=x}$ and $\widehat{N}(x):=\big(D'(w)/w\big)\big|_{w^2=x}$, and then independently verified by direct 60-digit numerical evaluation of the defining Verlinde sum (5) for every n tested.
In every case, writing $Q(\tau)=\sum_{k=0}^{K} q_k\tau^k$, the recursion is

$$V_n(m)=c_1V_{n-1}(m)+c_2V_{n-2}(m)+\cdots+c_KV_{n-K}(m),\qquad n>K,\qquad c_k:=-\frac{q_k}{q_0}.$$

We refer to the initial values $V_1(m)$,...,$V_K(m)$ as the seeds of the recursion.

Example 6.5. For $m=3$ $(K=1)$ we have

$$V_n(3)=V_{n-1}(3),\qquad n>1.$$

Seed: $V_1(3)=2$. This recovers previously obtained information.

Example 6.6. For $m=5$ $(K=6)$ we have

$$V_n(5)=45\,V_{n-1}-810\,V_{n-2}+7425\,V_{n-3}-36450\,V_{n-4}+91125\,V_{n-5}-91125\,V_{n-6},\qquad n>6.$$

Seed: $V_1(5),\dots,V_6(5)=90,\ 810,\ 8100,\ 85050,\ 911250,\ 9841500$.

Example 6.7. For $m=6$ $(K=10)$ we have

$$\begin{aligned}V_n(6)=\ &166\,V_{n-1}-11583\,V_{n-2}+446148\,V_{n-3}-10526031\,V_{n-4}+160022790\,V_{n-5}\\ &-1599105969\,V_{n-6}+10433249712\,V_{n-7}-42702347232\,V_{n-8}+99179645184\,V_{n-9}\\ &-99179645184\,V_{n-10},\qquad n>10.\end{aligned}$$

Seed: $V_1(6),\dots,V_{10}(6)=$

$$332,\ 8780,\ 288812,\ 10156940,\ 363507692,\ 13067079500,\ 470242412972,$$
$$16927176136460,\ 609364389580652,\ 21936992483887820.$$

Example 6.8. For $m=7$ $(K=15)$:

$$\begin{aligned}V_n(7)=\ &504\,V_{n-1}-108486\,V_{n-2}+13215447\,V_{n-3}-1023942465\,V_{n-4}+53971394715\,V_{n-5}\\ &-2018505657735\,V_{n-6}+54972824448402\,V_{n-7}-1105751293418835\,V_{n-8}\\ &+16497990847533951\,V_{n-9}-181643903852608890\,V_{n-10}+1452250517248048266\,V_{n-11}\\ &-8172324365150558151\,V_{n-12}+30585530792345653278\,V_{n-13}-68122318582951682301\,V_{n-14}\\ &+68122318582951682301\,V_{n-15},\qquad n>15.\end{aligned}$$

Seed: $V_1(7), \dots, V_{15}(7) =$

1008, 74088, 7279146, 760809672, 80448621498, 8525323959642, 903825853821702, 95828299059904488, 10160373167700798168, 1077275750776882216938, 114220583437370668330464, 12110495610281825876484330, 1284042709290383910109981746, 136143535334203397185360075278, 144349265771536836900009560927616.

7. Proof of Theorems 1.3 and 1.4

7.1. **Proof Theorem 1.3.** We now show that $D(w)$ is essentially the cube of a polynomial, which substantially improves the length of the recursion of Corollary 6.3.

For $(j_1, j_2, j_3) \in (\mathbb{Z}/m\mathbb{Z} \setminus \{0\})^3$ admissible values of ℓ for which $j_1 + j_2 + j_3 = \ell m$ are $\ell \in \{1, 2\}$.

Lemma 7.1. *The symmetric group on three letters S_3 acts on the set of nonzero triples $(j_1, j_2, j_3) \in A(m) \setminus A_0(m)$ by permutation, preserving Λ_m. Every orbit has size 1, 3, or 6. An orbit of size 1 requires $j_1 = j_2 = j_3$, so then $3j_1 \equiv 0 \pmod m$ with $j_1 \not\equiv 0$ except possibly only when $3 \mid m$. In this case there are exactly two such orbits, when $j_1 = m/3$ and $j_1 = 2m/3$.*

Proof. Each orbit has size equal to $6/|\mathrm{Stab}|$ where Stab is the order of the stabilizer of any point in the orbit. The stabilizer is all of S_3 precisely when $j_1 = j_2 = j_3$ (yielding the orbit size 1), a single transposition precisely when two indices coincide (yielding the orbit size 3), trivial otherwise (yielding the orbit size 6). For $j_1 = j_2 = j_3 =: j$, the constraint $3j \equiv 0 \pmod m$ has the excluded solution $j \equiv 0$ and, when $3 \mid m$, the further solutions are $j = m/3, 2m/3$. □

Lemma 7.2. *The function $f(x) := \log \sin x$ is strictly concave on $(0, \pi)$. Consequently, for $\theta_1, \theta_2, \theta_3 \in (0, \pi)$ with $\theta_1 + \theta_2 + \theta_3 = k\pi$ fixed ($k \in \{1, 2\}$),*

$$\sin\theta_1 \sin\theta_2 \sin\theta_3 \le \sin^3\Big(\frac{k\pi}{3}\Big), \tag{27}$$

with equality if and only if $\theta_1 = \theta_2 = \theta_3 = k\pi/3$. In particular, the maximum value of the left-hand-side of (27) is $3\sqrt{3}/8$ which is attained only at $\theta_1 = \theta_2 = \theta_3 = k\pi/3$, $k \in \{1, 2\}$. Hence, the eigenvalue $3\Lambda = (-1)^k 4 \prod_{\ell=1}^3 \sin\theta_\ell = \mp 3\sqrt{3}/2$ only comes from the singleton orbits of Lemma 7.1.

Proof. Clearly, $f''(x) = -\csc^2 x < 0$ on $(0, \pi)$, so f is strictly concave. Jensen's inequality gives

$$\tfrac{1}{3}(f(\theta_1) + f(\theta_2) + f(\theta_3)) \le f(\tfrac{\theta_1+\theta_2+\theta_3}{3}) = f(k\pi/3),$$

with equality precisely when $\theta_1 = \theta_2 = \theta_3$. Upon exponentiating gives the claim. At the maximum, we have that $j_1 = j_2 = j_3 = m/3$ (resp. $2m/3$) gives $\theta_i = \pi/3$ (resp. $2\pi/3$), and $3\Lambda = -4 \sin\theta_1 \sin\theta_2 \sin\theta_3$, and only at these two values $|3\Lambda|$ attains the strict global maximum over all real triples with angle sum $k\pi$, $k \in \{1, 2\}$. □

Proof of Theorem 1.3. By Lemma 7.1, $D(w) = \prod_{j=(j_1,j_2)}(w - r_j)$ is a product over all ordered nonzero triples $(j_1, j_2, j_3) \in A(m) \setminus A_0(m)$. When grouping by S_3-orbits, we can write

$$D(w) = \prod_{\text{orbits } O} (w - 3\Lambda_O)^{|O|},$$

where $\Lambda_O = \Lambda_m(j) = r_j/3$ for a representative $j = (j_1, j_2, j_3)$ of the orbit O.

By Lemma 7.2, the two singleton orbits exist only when $3 \mid m$, and the corresponding rescaled eigenvalues are $r = \pm 3\sqrt{3}/2$. Hence, the singleton orbits contribute the factor $w^2 - 27/4$. Every other orbit has size 3 or 6, hence multiplicity divisible by 3. We then can write the polynomial E as

$$E(w) = \prod_{\text{orbits } O \text{ of size } 3} (w - 3\Lambda_O) \prod_{\text{orbits } O \text{ of size } 6} (w - 3\Lambda_O)^2.$$

The degree count follows from $\deg D = (m-1)(m-2)$ taking into account when the exceptional factor of degree 2 is present when m is divisible by 3.

It is left to prove that $E(w) \in \mathbb{Q}[w]$. We proved that $E^3(w) = D(w)/H_m(w)$, where $H_m(w) \equiv 1$ if $3 \nmid m$ and $H_m(w) = w^2 - 27/4$, if $3 \mid m$. Since $D(w)/H_m(w)$ is monic and its monic cube root is unique, every $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ fixes E, hence $E(w) \in \mathbb{Q}[w]$. □

7.2. **Proof of Theorem 1.4.** From Theorem 1.3 we have that

$$w\frac{E'(w)}{E(w)} = \sum_{a\in\mathcal{E}} \frac{w\mu(a)}{w-a} = -\sum_{n\geq 1}\sum_{a\in\mathcal{E}_m} \mu(a)\left(\frac{w}{a}\right)^n.$$

The statement follows immediately from Theorem 1.2 and definition of $S_{2n}(m)$ and $V_0(m)$.

7.3. **Further possible reduction of the recursion length.**

7.3.1. *Impact on the recursion length.* Since D is even, so is E. By writing $\widehat{E}(v) := E(w)|_{w^2=v}$, the largest recursion length is comparable to the degree of $\widehat{E}$ in v. Specifically,

$$\text{recursion length} \leq \deg \widehat{E} + [3 \mid m] = \lfloor \frac{(m-1)(m-2)}{6} \rfloor + [3 \mid m].$$

This is a significant reduction when we only studied D, which, in case when $\mathrm{GCD}(P,Q)$ is trivial, gave a recursion length of $K = (m-1)(m-2)/2$. The gap between the current bound and the true minimal order goes to the question of whether or not $\widehat{E}$ itself has repeated roots.

Lemma 7.3. *The polynomial $\widehat{E}(v)$ is squarefree if and only if $m \in \{3,4,5\}$. Equivalently, if $m \geq 6$, then $D(w)$ always has a root of multiplicity at least 6.*

Proof. For $m = 3$, we have $E \equiv 1$, $E(w) = w^2 - 4$ for $m = 4$ and $E(w) = 16w^4 - 100w^2 + 125$ for $m = 5$. Their corresponding polynomials $\widehat{E}$ are obviously squarefree.

For $m \geq 6$, consider the triple $(1, 2, m-3)$. Its three coordinates are pairwise distinct mod m and none of the three coordinates is $\equiv 0 \pmod m$. So by the characterization of $A_0(m)$ as the triples in $A(m)$ with at least one zero coordinate, this triple lies in $A(m) \setminus A_0(m)$, i.e. its eigenvalue $3\Lambda_m(1,2,m-3)$ is nonzero. Since its three coordinates are pairwise distinct, its S_3-orbit has size exactly 6 by Lemma 7.1. As such, the E-multiplicity is at least 2 at $w = 3\Lambda_m(1,2,m-3)$. Hence $\widehat{E}$ is not squarefree for any $m \geq 6$. □

Lemma 7.4. *Let $N := \mathrm{lcm}(m,4)$. Every root of $D(w)$ lies in $\mathbb{Q}(\zeta_N)^+$, the maximal real subfield of $\mathbb{Q}(\zeta_N)$, of degree $\varphi(N)/2$ over $\mathbb{Q}$. Consequently $E(w)$ splits completely over $\mathbb{Q}(\zeta_N)^+$, and every irreducible factor of $E(w)$ has degree dividing $\varphi(N)/2$.*

Proof. Each root of $D(w)$ has the form $3\Lambda(j) = -(\sin\frac{2\pi j_1}{m} + \sin\frac{2\pi j_2}{m} + \sin\frac{2\pi j_3}{m})$. Since each term $\sin(2\pi j/m) = \frac{1}{2i}(\zeta_m^j - \zeta_m^{-j}) \in \mathbb{Q}(\zeta_m, i) = \mathbb{Q}(\zeta_{\mathrm{lcm}(m,4)})$, so does λ_j. Since $3\Lambda(j)$ is real, it lies in the fixed field of complex conjugation, $\mathbb{Q}(\zeta_N)^+$, of degree $\varphi(N)/2$. As $\mathbb{Q}(\zeta_N)$ is Galois over $\mathbb{Q}$, it is automatically a splitting field for the minimal polynomial of any of its elements. Moreover, the cyclotomic Galois group is abelian, complex conjugation is central, and therefore its fixed field is Galois over $\mathbb{Q}$. The tower law hence yields that every irreducible factor of $E(w)$ has degree dividing $\varphi(N)/2$. □

Remark 7.5. Every irreducible factor degree divides $\varphi(\mathrm{lcm}(m,4))/2$. The recursion length will shorten further if any of these irreducible factors are repeated.

7.4. **Examples of shorter recursion formulas.** In what we write below, Recursion A refers to writing G in terms of D'/D with further reduction of D. Recursion B comes from using Theorem 1.3. The formulas below can be compared to those in Section 6.3.

Example 7.6. Let $m = 5$. Recursion A length was $K = 6$; Recursion B has length $L = 2$. The recursion formula using Theorem 1.3 is the following. For $n > 2$, $V_n(5) = 15V_{n-1}(5) - 45V_{n-2}(5)$ with $V_1(5) = 90$ and $V_2(5) = 810$.

Example 7.7. Let $m = 6$. Recursion A length was $K = 10$; Recursion B has length $L = 3$. The recursion formula using Theorem 1.3 is the following. For $n > 3$:

$$V_n(6) = 49V_{n-1}(6) - 504V_{n-2}(6) + 1296V_{n-3}(6)$$

with

$$V_1(6) = 332,\ V_2(6) = 8780 \text{ and } V_3(6) = 288812.$$

Example 7.8. Let $m = 7$. Recursion A length was $K = 15$; Recursion B has length $L = 4$. The recursion formula using Theorem 1.3 is the following. For $n > 4$,

$$V_n(7) = 147V_{n-1}(7) - 4851V_{n-2}(7) + 55566V_{n-3}(7) - 194481V_{n-4}(7)$$

with

$$V_1(7) = 1008,\ V_2(7) = 74088,\ V_3(7) = 7279146 \text{ and } V_4(7) = 760809672.$$

References

[AMW01] Alekseev, A., Meinrenken, E., Woodward, C.: *The Verlinde formulas as fixed point formulas*, J. Symplectic Geom. **1** (2001), no. 1, 1–46. Correction: J. Symplectic Geom. 1 (2002), no. 2, 427–434.

[Be96] Beauville, A.: *Conformal blocks, fusion rules and the Verlinde formula*, in: Proceedings of the Hirzebruch 65 Conference on Algebraic Geometry (Ramat Gan, 1993), Israel Math. Conf. Proc., vol. 9, Bar-Ilan Univ., Ramat Gan, 1996, pp. 75–96.

[BL94] Beauville, A., and Laszlo, Y.: *Conformal blocks and generalized theta functions*, Comm. Math. Phys. **164** (1994), 385–419.

[BT93] Blau, M. and Thompson, G.: *Derivation of the Verlinde formula from Chern-Simons theory and the G/G model*, Nuclear Phys. B, **408** (1993), 345–390.

[Bo91] Bott, R.: *On E. Verlinde's formula in the context of stable bundles*, Internat. J. Modern Phys. A **6** (1991), no. 16, 2847–2858.

[BoVe09] Boysal, A., Vergne, M.: *Paradan's wall crossing formula for partition functions and Khovanski–Pukhlikov differential operator*, Ann. Inst. Fourier (Grenoble) **59** (2009), no. 5, 1715–1752.

[BrVe97] Brion, M., Vergne, M.: *Residue formulae, vector partition functions and lattice points in rational polytopes*, J. Amer. Math. Soc. **10** (1997), no. 4, 797–833.

[CHJSV23a] Cadavid, C., Hoyos, P., Jorgenson, J., Smajlović, L., and Velez, J.: *On an approach for evaluating certain trigonometric character sums using the discrete time heat kernel*, European J. Combin. **108** (2023), Paper No. 103635.

[Do92] Dowker, J. S.: *On Verlinde's formula for the dimensions of vector bundles on moduli spaces*, J. Phys. A: Math. Gen. **25** (1992), no. 9, 2641–2648.

[Fa94] Faltings, G.: *A proof for the Verlinde formula*, J. Algebraic Geom. **3** (1994), 347–374.

[JKS24] Jorgenson, J., Karlsson, A., and Smajlović, L.: *The resolvent kernel on the discrete circle and twisted cosecant sums*, J. Math. Anal. Appl. **538** (2024), 128454.

[KM26] Karlsson, A., Müller, D.: *A discrete approach to Dirichlet L-functions, their special values and zeros*, https://arxiv.org/abs/2512.01779.

[Ku22] Kumar, S.: *Conformal Blocks, Generalized Theta Functions and the Verlinde Formula*, Cambridge University Press, 2022.

[KNR94] Kumar, S., Narasimhan, M. S., Ramanathan, A.: *Infinite Grassmannians and moduli spaces of G-bundles*, Math. Ann. **300** (1994), no. 1, 41–75.

[LM22] Loizides, Y., Meinrenken, E.: *The decomposition formula for Verlinde sums*, Ann. Inst. Fourier (Grenoble) **72** (2022), no. 3, 1207–1248.

[LM22b] Loizides, Y., Meinrenken, E.: *Verlinde sums and* $[Q, R] = 0$, preprint (2022), 50 pp.

[Sc08] Schottenloher, M.: *A mathematical introduction to conformal field theory*, Lecture Notes in Physics, **759** (2008) Springer-Verlag, Berlin, pp. xvi+249.

[So96] Sorger, C.: *La formule de Verlinde*, Séminaire Bourbaki, Vol. 1994/95 Astérisque **237** (1996), Exp. No. 794, 3, 87–114.
[Sz91] Szenes, A.: *Verification of Verlinde's formulas for SU(2)*, Int. Math. Res. Not. 1991 (1991), no. 7, 93–98.
[Sz95] Szenes, A.: *The combinatorics of the Verlinde formula*, in: Vector bundles in algebraic geometry (Durham, 1993), London Mathematical Society Lecture Note Series, vol. 208, Cambridge University Press, 1995, 241–253.
[Sz98] Szenes, A.: *Iterated residues and multiple Bernoulli polynomials*, Int. Math. Res. Not. **18** (1998), 937–956.
[Sz03] Szenes, A.: *Residue theorem for rational trigonometric sums and Verlinde's formula*, Duke Math. J. **118** (2003), no. 2, 189–227.
[SV03] Szenes, A., Vergne, M.: *Residue formulae for vector partitions and Euler–Maclaurin sums*, Adv. Appl. Math. **30** (2003), no. 1-2, 295–342.
[Th94] Thaddeus, M.: *Stable pairs, linear systems and the Verlinde formula*, Invent. Math. **117** (1994), no. 2, 317–353.
[TUY89] Tsuchiya, A., Ueno, K., and Yamada, Y.: *Conformal field theory on universal family of stable curves with gauge symmetries*, Adv. Studies in Pure Math. **19** (1989), 459–565.
[Ve03] Vergne, M.: *Residue formulae for Verlinde sums, and for number of integral points in convex rational polytopes*, Mezzetti, Emilia (ed.) et al., European women in mathematics. Proceedings of the tenth general meeting, (EWM'2001), Malta, August 24–30, 2001, River Edge, NJ: World Scientific (ISBN 981-238-190-2/hbk), 225–285, 2003.
[Ve88] Verlinde, E.: *Fusion rules and modular transformations in 2d conformal field theory*, Nucl. Phys. B **300** (1988), 360–376.
[Za96] Zagier, D.: *Elementary aspects of the Verlinde formula and of the Harder–Narasimhan–Atiyah–Bott formula*, Israel Math. Conf. Proc. **9** (1996), 445–462.

Appendix

Let us now present all information for various sets of Verlinde sums for fixed m.

For $m = 5, 6, 7, 9, 10, 14$ each subsection records the following computations: (a) the spectral determinant polynomial $D(w)$; (b) the resolvent kernel $\mathcal{G}(w) = D'(w)/D(w)$, in reduced closed form; (c) the factorization of $D(w)$, with every root exhibited in closed radical form, and the resulting expression of $V_n(m)$ as an explicit finite sum of n-th powers, meaning the associated Binet formula. For the cases $m = 5, 6, 7$, the data repeats some results from above.

$m = 5$.

The denominator.

$$D(w) = \frac{(16w^4 - 100w^2 + 125)^3}{4096}.$$

The resolvent kernel. With $E(w) := 16w^4 - 100w^2 + 125$, so that $D = E^3/4096$, Theorem 1.3 gives $\mathcal{G}(w) = D'(w)/D(w) = 3E'(w)/E(w)$:

$$\mathcal{G}(w) = \frac{192w^3 - 600w}{16w^4 - 100w^2 + 125}.$$

Factorization and the Verlinde sum. As E is even in w, the substitution $x = w^2$ reduces it to the quadratic $16x^2 - 100x + 125 = 0$, with roots

$$x = \frac{25 \pm 5\sqrt{5}}{8}.$$

The four roots of $E(w)$ are therefore

$$w = \pm\sqrt{\frac{25 + 5\sqrt{5}}{8}}, \qquad w = \pm\sqrt{\frac{25 - 5\sqrt{5}}{8}},$$

each of D-multiplicity 3. Setting $y := \frac{3}{4} \cdot 25/x = 75/(4x)$ at each of the two values of x gives $y = \frac{3(5\mp\sqrt{5})}{2}$, and for $n \geq 1$ we have

$$V_n(5) \;=\; 6\left[\left(\frac{3(5+\sqrt{5})}{2}\right)^n + \left(\frac{3(5-\sqrt{5})}{2}\right)^n\right] = \begin{cases} 2\cdot 3^{n+1}5^{\frac{n+1}{2}}F_n, & n \text{ odd} \\ 2\cdot 3^{n+1}5^{\frac{n}{2}}(F_{n-1}+F_{n+1}), & n \text{ even} \end{cases}, \quad (28)$$

where F_n denotes the nth Fibonacci number. The sum of coefficients is $6+6 = 12 = (5-1)(5-2)$.

$m = 6$.

The denominator. Since $3 \mid 6$. We have that

$$D(w) = \left(w^2 - \frac{27}{4}\right)(w^2-3)^6\left(w^2-\frac{3}{4}\right)^3.$$

The resolvent kernel. Summing logarithmic derivatives with D-multiplicities $1, 6, 3$, yielding

$$\mathcal{G}(w) = 2w\left[\frac{1}{w^2-\frac{27}{4}} + \frac{6}{w^2-3} + \frac{3}{w^2-\frac{3}{4}}\right] = \frac{320w^5 - 2496w^3 + 2988w}{16w^6 - 168w^4 + 441w^2 - 243}.$$

Factorization and the Verlinde sum. Every root is an elementary square root: $w = \pm\sqrt{3}$ (multiplicity 6), $w = \pm\sqrt{3}/2$ (multiplicity 3), and the exceptional pair $w = \pm\frac{3\sqrt{3}}{2}$ (multiplicity 1, the singleton S_3-orbit of Lemma 7.1). With $x = w^2 \in \{3, \frac{3}{4}, \frac{27}{4}\}$ and $y = 27/x$,

$$V_n(6) \;=\; 2\cdot 4^n + 12\cdot 9^n + 6\cdot 36^n, \qquad n \geq 1.$$

The sum of coefficients is $2 + 12 + 6 = 20 = (6-1)(6-2)$.

The recursion. For $n > 3$

$$V_n(6) = 49V_{n-1}(6) - 504V_{n-2}(6) + 1296V_{n-3}(6)$$

with the seeds $V_1(6) = 332$, $V_2(6) = 8780$ and $V_3(6) = 288812$.

$m = 7$.

The denominator.

$$D(w) = \frac{(4w^2-7)^6(64w^6 - 560w^4 + 1176w^2 - 343)^3}{2^{30}}.$$

The resolvent kernel. With $E(w) := \frac{1}{1024}(4w^2-7)^2(64w^6 - 560w^4 + 1176w^2 - 343)$, so that $D = E^3$,

$$\begin{aligned}\mathcal{G}(w) = 3\,\frac{E'(w)}{E(w)} &= \frac{48w}{4w^2-7} \;+\; \frac{1152w^5 - 6720w^3 + 7056w}{64w^6 - 560w^4 + 1176w^2 - 343} \\ &= \frac{7680w^7 - 61824w^5 + 131712w^3 - 65856w}{256w^8 - 2688w^6 + 8624w^4 - 9604w^2 + 2401}.\end{aligned}$$

Factorization and the Verlinde sum. $E(w)$ splits into an elementary quadratic-in-x factor and an irreducible cubic-in-x factor:

$$E(w) = \tfrac{1}{1024}(4w^2-7)^2 \cdot (64w^6 - 560w^4 + 1176w^2 - 343).$$

The first factor gives the elementary pair $w = \pm\sqrt{7}/2$, matching the eigenvalue of D-multiplicity 6 recorded in Example 3.4. The second, in $x = w^2$, is the irreducible cubic $64x^3-560x^2+1176x-343 = 0$; it has three real roots, given in closed trigonometric form by

$$x_k = \frac{35}{12} + \frac{7\sqrt{7}}{6}\cos\left(\frac{1}{3}\arccos\frac{\sqrt{7}}{14} - \frac{2\pi k}{3}\right), \qquad k = 0,1,2, \tag{29}$$

matching the three magnitude classes of Example 3.4. Equivalently, in radical form: with $\omega := e^{2\pi i/3}$ and

$$C_\pm := \sqrt[3]{\frac{2401}{3456}\left(1 \pm 3\sqrt{3}\,i\right)}$$

and

$$x_k = \frac{35}{12} + \omega^k C_+ + \omega^{-k} C_-, \qquad k = 0,1,2, \tag{30}$$

taking C_+, C_- to be complex conjugate cube roots of the (complex-conjugate) radicands above; the two formulas agree since $C_\pm = \frac{7\sqrt{7}}{12}e^{\pm i\theta}$ with $\theta = \frac{1}{3}\arccos(\sqrt{7}/14)$, so $\omega^k C_+ + \omega^{-k}C_- = 2\,\mathrm{Re}(\omega^k C_+) = \frac{7\sqrt{7}}{6}\cos(\theta - \frac{2\pi k}{3})$. The six roots of this factor are $w = \pm\sqrt{x_k}$, $k = 0,1,2$, each of D-multiplicity 3.

Setting $y_k := \frac{3}{4}\cdot 49/x_k = 147/(4x_k)$, the roots of the reciprocal cubic $y^3-126y^2+2205y-9261 = 0$, gives

$$V_n(7) \;=\; 12\cdot 21^n \;+\; 6\sum_{k=0}^{2} y_k^n, \qquad y_k = \frac{147}{4x_k}, \quad x_k \text{ as in (30)}.$$

The sum of coefficients is $12 + 6\cdot 3 = 30 = (7-1)(7-2)$.

$m = 9$.

The denominator. Here again $3 \mid 9$. Writing

$$P(w) := 64w^6 - 288w^4 + 324w^2 - 27, \qquad R(w) := 64w^6 - 432w^4 + 648w^2 - 243,$$

one has, exactly,

$$D(w) = 2^{-54}\left(w^2 - \frac{27}{4}\right)P(w)^3 R(w)^6,$$

a monic polynomial of degree $2 + 18 + 36 = 56 = (9-1)(9-2)$.

The resolvent kernel. Summing logarithmic derivatives with D-multiplicities $1, 3, 6$:

$$\mathcal{G}(w) = 2w\left[\frac{1}{w^2 - \frac{27}{4}}\right] + 3\frac{P'(w)}{P(w)} + 6\frac{R'(w)}{R(w)},$$

equivalently, as a single reduced fraction, $\mathcal{G}(w) = f(w)/g(w)$ where

$$f(w) = 917504w^{13} - 14524416w^{11} + 83109888w^9 - 216870912w^7 + 265472640w^5 - 135768960w^3 + 18475776w$$

and

$$\begin{aligned} g(w) = {} & 16384w^{14} - 294912w^{12} + 1990656w^{10} - 6414336w^8 + 10450944w^6 \\ & - 8258112w^4 + 2624400w^2 - 177147. \end{aligned}$$

Factorization and the Verlinde sum. The exceptional pair is again $w = \pm\frac{3\sqrt{3}}{2}$ (D-multiplicity 1). The remaining 54 roots split into two Galois orbits of triples under $j \mapsto 4j \pmod 9$ — the order-3 subgroup of $(\mathbb{Z}/9\mathbb{Z})^\times$ generating the cyclic cubic subfield of $\mathbb{Q}(\zeta_{36})^+$ relevant here — corresponding exactly to the factors P, R. In $x = w^2$ these are the two cubics

$$64x^3 - 288x^2 + 324x - 27 = 0 \qquad \text{(roots of } P\text{, } D\text{-multiplicity 3 each)},$$

and

$$64x^3 - 432x^2 + 648x - 243 = 0 \qquad \text{(roots of } R\text{, } D\text{-multiplicity 6 each)}.$$

Unlike the $m = 7$ case, both cubics have Cardano angle equal to a rational multiple of π, and the relevant root of unity here is indexed by m itself, thus giving the closed forms

$$x_k^P = \frac{3}{2}\left(1 + \cos\Big(\frac{2\pi}{9} - \frac{2\pi k}{3}\Big)\right), \qquad k = 0, 1, 2, \tag{31}$$

$$x_k^R = \frac{9}{4} + \frac{3\sqrt{3}}{2}\cos\Big(\frac{\pi}{18} - \frac{2\pi k}{3}\Big), \qquad k = 0, 1, 2. \tag{32}$$

The full set of 54 nonexceptional roots of $D(w)$ is $w = \pm\sqrt{x_k^P}$ (multiplicity 3) and $w = \pm\sqrt{x_k^R}$ (multiplicity 6), $k = 0, 1, 2$.

Setting $y_k^P := \frac{3}{4} \cdot 81/x_k^P = 243/(4x_k^P)$ and likewise $y_k^R = 243/(4x_k^R)$,

$$V_n(9) \;=\; 2 \cdot 9^n \;+\; 6\sum_{k=0}^{2}\big(y_k^P\big)^n \;+\; 12\sum_{k=0}^{2}\big(y_k^R\big)^n, \tag{33}$$

with x_k^P, x_k^R as above. In particular

$$V_1(9) = 6336, \qquad V_2(9) = 2873880, \qquad V_3(9) = 1832410026.$$

The sum of coefficients is $2 + 6 \cdot 3 + 12 \cdot 3 = 56 = (9-1)(9-2)$.

We now proceed with two final examples: $m = 10$ and $m = 14$.

$m = 10$.

The denominator. Again $3 \nmid 10$, so $D = E^3$:

$$D(w) = \frac{\big(w^4 - 5w^2 + 5\big)^6\big(16w^4 - 100w^2 + 5\big)^3\big(16w^4 - 100w^2 + 125\big)^3\big(16w^4 - 20w^2 + 5\big)^6}{2^{48}},$$

monic of degree $72 = (10-1)(10-2)$. Notably, the factor $16w^4 - 100w^2 + 125$ is *identical* to the $m = 5$ denominator quartic. This is expected because of the presence of the terms for $m = 5$ in the series for $m = 10$.

The resolvent kernel.

$$\mathcal{G}(w) = \frac{294912w^{15} - 4915200w^{13} + 31150080w^{11} - 95232000w^9 + 147686400w^7 - 112080000w^5 + 35649000w^3 - 2887500w}{4096w^{16} - 76800w^{14} + 560640w^{12} - 2040000w^{10} + 3951200w^8 - 4057500w^6 + 2042625w^4 - 403125w^2 + 15625}.$$

Factorization and the Verlinde sum. Passing to $x = w^2$, each quartic factor of D becomes a quadratic in x, all four solvable directly:

$$\widehat{E}_b(x) = \frac{(x^2 - 5x + 5)^2(16x^2 - 100x + 5)(16x^2 - 100x + 125)(16x^2 - 20x + 5)^2}{65536},$$

with roots

$$x = \frac{5 \pm \sqrt{5}}{2} \text{ (mult. 2)}, \qquad x = \frac{25 \pm 11\sqrt{5}}{8} \text{ (mult. 1)},$$

$$x = \frac{25 \pm 5\sqrt{5}}{8} \text{ (mult. 1, the embedded } m = 5 \text{ pair)}, \qquad x = \frac{5 \pm \sqrt{5}}{8} \text{ (mult. 2)}.$$

With $c_{10} = 3 \cdot 10^2/4 = 75$ and $y := c_m/x$,

$$y\Big(\tfrac{5\pm\sqrt{5}}{2}\Big) = \tfrac{75\mp 15\sqrt{5}}{2}, \qquad y\Big(\tfrac{25\pm 11\sqrt{5}}{8}\Big) = 750 \mp 330\sqrt{5},$$
$$y\Big(\tfrac{25\pm 5\sqrt{5}}{8}\Big) = 30 \mp 6\sqrt{5}, \qquad y\Big(\tfrac{5\pm\sqrt{5}}{8}\Big) = 150 \mp 30\sqrt{5},$$

giving

$$V_n(10) = 6\Big\{\ 2\Big[\Big(\tfrac{75-15\sqrt{5}}{2}\Big)^n + \Big(\tfrac{75+15\sqrt{5}}{2}\Big)^n\Big] + \big(750 - 330\sqrt{5}\big)^n + \big(750 + 330\sqrt{5}\big)^n \tag{34}$$
$$+ \big(30 - 6\sqrt{5}\big)^n + \big(30 + 6\sqrt{5}\big)^n + 2\Big[\big(150 - 30\sqrt{5}\big)^n + \big(150 + 30\sqrt{5}\big)^n\Big]\Big\}, \quad n \geq 1.$$

The sum of coefficients is $6(2 + 2 + 1 + 1 + 1 + 1 + 2 + 2) = 72$, as required. The middle pair, $(30 \mp 6\sqrt{5})^n$, is exactly 4^n times the $m = 5$ contribution $y = 3(5 \mp \sqrt{5})/2$ of (10.1) — the embedding rescales by $(c_{10}/c_5) = (10/5)^2 = 4$, as it must. Explicitly,

$$V_1(10) = 13860, \qquad V_2(10) = 13985460, \qquad V_3(10) = 19896143400.$$

The recursion. For $n > 8$,

$$\begin{aligned} V_n(10) = {} & 1935V_{n-1}(10) - 735345V_{n-2}(10) + 109552500V_{n-3}(10) - 8001180000V_{n-4}(10) \\ & + 309825000000V_{n-5}(10) - 6386040000000V_{n-6}(10) \\ & + 65610000000000V_{n-7}(10) - 262440000000000V_{n-8}(10) \end{aligned}$$

with $V_1(10), \ldots, V_8(10)$ given by 13860, 13985460, 19896143400, 29434232085300, 43760418628432 500, 65103806075857659000, 96866509540005589702500, 144127568255155203219232500.

$m = 14$**: an embedded cube root.** Computations with $m = 10$ showed a *square-root* example reappearing inside a new level, $m = 14 = 2 \cdot 7$ shows the same phenomenon one step up.

The denominator. Since $3 \nmid 14$, $D = E^3$ with no exceptional factor. We find

$$D(w) = \frac{(4w^2-7)^6\big(w^6-7w^4+14w^2-7\big)^6\big(64w^6-560w^4+952w^2-7\big)^3\big(64w^6-560w^4+1176w^2-343\big)^3\big(64w^6-336w^4+140w^2-7\big)^6\big(64w^6-112w^4+56w^2-7\big)^6}{2^{120}},$$

a monic polynomial of degree $156 = (14-1)(14-2)$.

Two of these six factors are *exactly* the two factors of $D(w)$ for $m = 7$ itself : $4w^2 - 7$ and $64w^6 - 560w^4 + 1176w^2 - 343$, unchanged, down to the coefficients. The other four factors are genuinely new to $m = 14$.

Verlinde sums. It is worth having $V_n(7)$ itself as a single self-contained formula into one line, so the cubic radicals are visible directly rather than assembled from two separate equations. With

$$\omega := e^{2\pi i/3}, \qquad C_\pm := \sqrt[3]{\frac{2401}{3456}\Big(1 \pm 3\sqrt{3}\,i\Big)},$$

we have

$$V_n(7) \ = \ 12 \cdot 21^n \ + \ 6 \cdot \left(\frac{147}{4}\right)^n \sum_{k=0}^{2} \left(\frac{35}{12} + \omega^k C_+ + \omega^{-k} C_-\right)^{-n}, \qquad n \geq 1.$$

The imaginary parts cancel term by term, since the three k-values run through a full conjugate orbit, thus leaving an integer for every n. This was checked directly at $n = 1, 2, 3$, reproducing $V_1(7) = 1008$, $V_2(7) = 74088$, $V_3(7) = 7279146$ to 50-digit precision with the imaginary part vanishing to that same precision.

Passing to $x = w^2$, the embedded cubic $64x^3 - 560x^2 + 1176x - 343$ has exactly the three roots $x_k = \frac{35}{12} + \omega^k C_+ + \omega^{-k} C_-$ appearing inside the formula above.

At level 14, $c_{14} = 3 \cdot 14^2/4 = 147$, so $y_k := c_{14}/x_k = 147/x_k$ — exactly 4 times the level-7 value $147/(4x_k)$ appearing in the boxed formula above, since $c_{14}/c_7 = (14/7)^2 = 4$. The embedded quadratic factor $4x-7$ contributes its single rescaled root the same way. Combining both embedded pieces, the *entire* embedded contribution collapses to $4^n\, V_n(7)$. The whole $m = 7$ spectrum reappears inside $m = 14$, rescaled by exactly $(c_{14}/c_7)^n = 4^n$.

The remaining four factors of $E(w)$, each new to $m = 14$, are, in $x = w^2$, with their E-multiplicities:

$$x^3 - 7x^2 + 14x - 7 \quad \text{(mult. 2)}, \qquad 64x^3 - 560x^2 + 952x - 7 \quad \text{(mult. 1)},$$
$$64x^3 - 336x^2 + 140x - 7 \quad \text{(mult. 2)}, \qquad 64x^3 - 112x^2 + 56x - 7 \quad \text{(mult. 2)}.$$

Solving each gives that

cubic	shift	$C_\pm$
$x^3 - 7x^2 + 14x - 7$	7/3	$\sqrt[3]{\frac{7}{54}(-1 \pm 3\sqrt{3}\,i)}$
$64x^3 - 560x^2 + 952x - 7$	35/12	$\sqrt[3]{\frac{10969}{3456} \pm \frac{3913\sqrt{3}}{1152}\,i}$
$64x^3 - 336x^2 + 140x - 7$	7/4	$\sqrt[3]{\frac{7}{2}\left(1 \pm \frac{\sqrt{3}}{9}i\right)}$
$64x^3 - 112x^2 + 56x - 7$	7/12	$\sqrt[3]{\frac{7}{3456}(-1 \pm 3\sqrt{3}\,i)}$

Writing $C_\pm^{(1)}, \ldots, C_\pm^{(4)}$ for the four pairs above in order and $C_\pm^{(0)}$ for $m = 7$'s own pair of $m = 7$ subsection of this Appendix, every root of $D(w)$ is accounted for, and $V_n(14)$ is, in full:

$$\begin{aligned} V_n(14) \;&= 4^n \left(12 \cdot 21^n + 6 \sum_{k=0}^{2} \left[\frac{147}{4\left(\frac{35}{12} + \omega^k C_+^{(0)} + \omega^{-k} C_-^{(0)}\right)} \right]^n \right) \\ &\quad + 12 \sum_{k=0}^{2} \left[\frac{147}{\frac{7}{3} + \omega^k C_+^{(1)} + \omega^{-k} C_-^{(1)}} \right]^n + 6 \sum_{k=0}^{2} \left[\frac{147}{\frac{35}{12} + \omega^k C_+^{(2)} + \omega^{-k} C_-^{(2)}} \right]^n \\ &\quad + 12 \sum_{k=0}^{2} \left[\frac{147}{\frac{7}{4} + \omega^k C_+^{(3)} + \omega^{-k} C_-^{(3)}} \right]^n + 12 \sum_{k=0}^{2} \left[\frac{147}{\frac{7}{12} + \omega^k C_+^{(4)} + \omega^{-k} C_-^{(4)}} \right]^n \end{aligned}$$

This was verified directly against 100-digit evaluation of the defining sum (5), with the imaginary parts cancelling to that same precision at every n tested:

$$V_1(14) = 176904, \quad V_2(14) = 2466167256,$$
$$V_3(14) = 47523549747996, \quad V_4(14) = 942433121112340536.$$

The recursion. For $n > 16$,

$$\begin{aligned} V_n(14) &= 24990 V_{n-1}(14) - 110391561 V_{n-2}(14) + 191178833013 V_{n-3}(14) \\ &\quad - 173643684876648 V_{n-4}(14) + 94555439941369680 V_{n-5}(14) - 331783566643614798 72 V_{n-6}(14) \\ &\quad + 78284473193712154199 04 V_{n-7}(14) - 12749897167572262175170 56 V_{n-8}(14) \\ &\quad + 14558624266917748801067827 2 V_{n-9}(14) - 11741021358725899855634890752 V_{n-10}(14) \\ &\quad + 66805218172764373589780948582 4 V_{n-11}(14) - 26513463184439223539851287330816 V_{n-12}(14) \\ &\quad + 71480247743288311983558821412864 0 V_{n-13}(14) \\ &\quad - 12426472330422917072242825639231488 V_{n-14}(14) \\ &\quad + 12518354271910264159702150409512550 4 V_{n-15}(14) \\ &\quad - 55344303096866431021841086021002854 4 V_{n-16}(14) \end{aligned}$$

with $V_1(14), \ldots, V_{16}(14)$ given by

176904, 2466167256, 47523549747996, 942433121112340536, 18750635619256136231244, 373213726101841157028558972, 7428848048726501458177459312788, 14787275707193555179729665376648464, 2943440256373791257710737826850288856744, 5858984189330963539226265956148373530112671 6, 11662440274746631858732019790125214694727893342 32, 2321435064504342238098319130857934660586731737 6979580, 462086890301572161379915294364229256198446851400244101276, 9197943869263408950920409929236677298908462024870299935144644, 1830871491879063501660614466632288358750498416429532820302140342 16, 3644391037195382365141804194836781078180376922894241912672127300912440.

If we had used the polynomial D itself for the recursion for $m = 14$, the formula would require 78 terms and the largest coefficient has 175 digits.

Jay Jorgenson
Department of Mathematics
The City College of New York
Convent Avenue at 138th Street
New York, NY 10031 U.S.A.
e-mail: jjorgenson@mindspring.com

Anders Karlsson
Section de mathématiques
Université de Genève
Case Postale 64, 1211
Genève 4, Suisse
e-mail: anders.karlsson@unige.ch
and
Matematiska institutionen
Uppsala universitet
Box 524, 751 20
Uppsala, Sweden
e-mail: anders.karlsson@math.uu.se

Lejla Smajlović
Department of Mathematics and Computer Science
University of Sarajevo
Zmaja od Bosne 35, 71 000 Sarajevo
Bosnia and Herzegovina
e-mail: lejlas@pmf.unsa.ba